

# Probability Guided Chemical Reaction Scopes - Addendum

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# 1 General

This addendum to the SI is meant to give the information on the analysis made to achieve the best model and to better understand the importance of each feature used by the model. The SI includes a list of the 10 best models that were extracted for each case and, with the best model chosen marked in green. Afterwards an analysis of the best performing model will be given, with the confusion matrix and the table of probabilities for each molecule. Those analysis are performed on each part in the model, rather it is the training, test or external validations. Each part will have a title indicating what is the attribution of the table. To understand how each feature contributes to the model, we performed the same analysis of the model, where each time we remove a different feature. The changes in accuracies and recall percentage indicate how the feature influences the classification.

## 2 Best model Specifications

### 2.1 Organocatalytic deuteration of aromatic aldehydes

The following sections contain the list of best models as extracted from the code in R provided in our GitHub account. The scopes and reaction details are given in the main SI. The reaction to which the following model attributes is as follows:

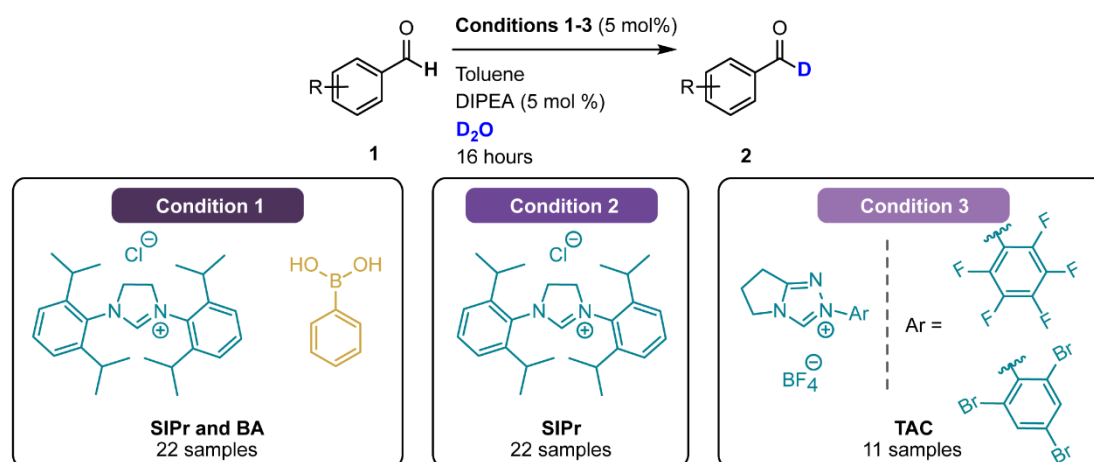
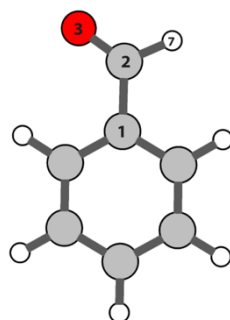


Figure S1: Representation of the reaction

### 2.1.1 Features extracted for model identification



| Feature                                        | Atom indices                                                                                            | Dataset naming scheme            | Manuscript naming scheme                     |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------|
| <b>Sterimol (along)</b>                        | 2-1                                                                                                     | B1, B5, L                        | $B_{1(2-1)}$ or $L_{2-1}$                    |
| <b>NPA charges</b>                             | 1 2 3 10                                                                                                | NPA_10                           | $q_1$                                        |
| <b>NPA charge differences</b>                  | 2-3 & 2-7                                                                                               | Diff_2_3                         | $\Delta q_{2-3}$<br>$\Delta q_{C=O}$         |
| <b>Dipole moment and its components</b>        | Origin at atom 1, y axis in the direction of 2, xy plane on the ring, z perpendicular to the ring plane | Total<br>dip_x<br>dip_y<br>dip_z | $\mu_{tot}$<br>$\mu_x$<br>$\mu_y$<br>$\mu_z$ |
| <b>Stretching vibration frequency</b>          | 2-3 & 2-7                                                                                               | -2-3-<br>-2-7-                   | $\nu_{C=O}$<br>$\nu_{C-H}$                   |
| <b>Bond lengths and distance between atoms</b> | 1-2 & 2-3 & 2-7                                                                                         | Dist(1, 2) or<br>Dist(2, 7)      | $d_{C-C}$<br>$d_{C-H}$                       |

**Figure S2 and Table S2:** Representation of the atom indexing and feature set

The dipole moment components are the decomposition of the total dipole ( $\mu_{Tot}$ ) with respect to a constant xyz coordinate system across all substrates. The *ipso* carbon on the benzaldehyde is set as the origin of the coordinate system, the y axis ( $\mu_y$ ) is defined from the *ipso* carbon towards the carbonyl carbon, the x axis ( $\mu_x$ ) is perpendicular to it on the plane of the benzene ring and the z axis ( $\mu_z$ ) is perpendicular to the plane of the ring.

### 2.1.2 Best model and model identification reports

Below is the data for the model presented in the main text as it appears in the output generated by our statistical code.

The top 10 models and their analyses as they resulted from the model identification code can be found on the project page on GitHub – [Top 10 models Deuteration](#)

The top 10 non-ordinal models and their analyses as they resulted from the model identification code can be found on the project page on GitHub – [Top 10 non-ordinal Deuteration](#)

The top 10 non-ordinal models using full, untruncated datasets and their analyses as they resulted from the model identification code can be found on the project page on GitHub – [Top 10 full data non-ordinal Deuteration](#)

The features in the most predictive model were the stretching vibration of the aldehydic carbon and oxygen, **-2-3-**; the distance between the aldehydic carbon and hydrogen, **Dist(2, 7)**; the component of the dipole moment in the direction going from the *ipso* carbon toward the aldehydic carbon, **dip\_y**; and the NPA charge of the *ipso* carbon, **NPA\_1** (Figure 3b in the main text). This classification model resulted in an accuracy of 91% for the training set, an accuracy of 92% for the test set (representing one misclassification out of thirteen examples), and an external validation of 100% accuracy. The model also provides the probability that each substrate would fall under each of the three reaction condition categories (Figure 3d in the main text).

These are the top-10 fitted models, as they result from the brute-force model search. In this case, the first-ranked model provided the best testing result and was therefore, further analyzed.

Below are the confusion matrices and probability heatmaps for the training, test, external validation, and prediction sets, as they result from our code. Next to each of them, there is a link that takes you back to the scope, with which you should be able to move back and forth.

| formula                                           | McFadden R2  |
|---------------------------------------------------|--------------|
| <b>class ~ -2-3- + dip_y + Dist(2, 7) + NPA_1</b> | <b>0.577</b> |
| class ~ -2-3- + dip_y + NPA_1 + NPA_3             | 0.577        |
| class ~ dip_y + NPA_3 + diff 1 2 + L              | 0.570        |
| class ~ dip_y + Dist(2, 7) + NPA_3 + diff 1 2     | 0.567        |

| formula                                       | McFadden R2 |
|-----------------------------------------------|-------------|
| class ~ -2-3- + dip_y + NPA_1 + NPA_2         | 0.557       |
| class ~ -2-3- + dip_y + Dist(2, 7) + diff 1 2 | 0.555       |
| class ~ -2-7- + dip_y + NPA_1 + NPA_3         | 0.555       |
| class ~ dip_y + NPA_1 + NPA_3 + diff 2 7      | 0.553       |
| class ~ dip_y + NPA_3 + NPA_7 + diff 1 2      | 0.551       |
| class ~ -2-3- + Total + NPA_1 + NPA_3         | 0.549       |

### Published Model

Full Model Stats - Overall Accuracy and Pseudo-R2

| Accuracy | McFadden_R2 |
|----------|-------------|
| 90.48%   | 0.577       |

Model Coefficients

|            | x       |
|------------|---------|
| -2-3-      | -0.380  |
| dip_y      | -1.568  |
| Dist(2, 7) | 295.849 |
| NPA_1      | 47.564  |

[Back to scope – Training Set](#)

## Training Set

### Confusion Matrix

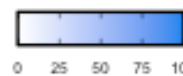
|       |               |               |              |               |
|-------|---------------|---------------|--------------|---------------|
| 1     | 12<br>(92.3%) | 1<br>(7.7%)   | 0<br>(0%)    | 13<br>(31%)   |
| 2     | 0<br>(0%)     | 17<br>(94.4%) | 1<br>(5.6%)  | 18<br>(42.9%) |
| 3     | 0<br>(0%)     | 2<br>(18.2%)  | 9<br>(81.8%) | 11<br>(26.2%) |
| total | 12<br>(100%)  | 20<br>(85%)   | 10<br>(90%)  | 42<br>(90.5%) |
|       | 1             | 2             | 3            | total         |

■ Accuracy ■ False ■ Precision ■ Size ■ True

1. 1st Place

## Probability Heatmap

|     | 1   | 2  | 3   | Exp |
|-----|-----|----|-----|-----|
| 2A  | 57  | 42 | 1   | 1   |
| 2G  | 98  | 2  | 0   | 1   |
| 2H  | 98  | 2  | 0   | 1   |
| 2L  | 88  | 11 | 0   | 1   |
| 2M  | 89  | 11 | 0   | 1   |
| 2O  | 100 | 0  | 0   | 1   |
| 2R  | 13  | 80 | 6   | 1   |
| 2S  | 71  | 28 | 0   | 1   |
| 2T  | 74  | 25 | 0   | 1   |
| 2U  | 99  | 1  | 0   | 1   |
| 2V  | 82  | 18 | 0   | 1   |
| 2J  | 87  | 13 | 0   | 1   |
| 2P  | 98  | 2  | 0   | 1   |
| 2Y  | 2   | 68 | 30  | 2   |
| 2Z  | 2   | 65 | 33  | 2   |
| 2AA | 1   | 51 | 48  | 2   |
| 2AE | 10  | 81 | 8   | 2   |
| 2AF | 46  | 53 | 1   | 2   |
| 2AG | 18  | 77 | 4   | 2   |
| 2AH | 16  | 79 | 5   | 2   |
| 2AJ | 2   | 62 | 36  | 2   |
| 2AK | 6   | 81 | 13  | 2   |
| 2AM | 7   | 81 | 13  | 2   |
| 2AO | 23  | 74 | 3   | 2   |
| 2W  | 27  | 70 | 3   | 2   |
| 2X  | 16  | 79 | 5   | 2   |
| 2AB | 3   | 71 | 26  | 2   |
| 2AC | 5   | 79 | 15  | 2   |
| 2AP | 20  | 76 | 4   | 2   |
| 2AQ | 22  | 75 | 4   | 2   |
| 2AR | 0   | 24 | 76  | 2   |
| 2AS | 1   | 44 | 55  | 3   |
| 2AT | 1   | 44 | 55  | 3   |
| 2AU | 38  | 60 | 2   | 3   |
| 2AV | 1   | 46 | 54  | 3   |
| 2AW | 5   | 79 | 16  | 3   |
| 2AX | 0   | 2  | 98  | 3   |
| 2AY | 0   | 23 | 77  | 3   |
| 2AZ | 0   | 2  | 98  | 3   |
| 2BA | 0   | 0  | 100 | 3   |
| 2BB | 0   | 0  | 100 | 3   |
| 2BC | 0   | 0  | 100 | 3   |



1. 1st Place

[Back to scope – Test Set](#)

Test Set

Confusion Matrix

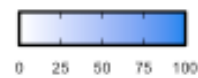
|       |             |             |           |               |
|-------|-------------|-------------|-----------|---------------|
| 1     | 9<br>(100%) | 0<br>(0%)   | 0<br>(0%) | 9<br>(69.2%)  |
| 2     | 1<br>(25%)  | 3<br>(75%)  | 0<br>(0%) | 4<br>(30.8%)  |
| 3     | 0<br>(0%)   | 0<br>(0%)   | 0<br>(0%) | 0<br>(0%)     |
| total | 10<br>(90%) | 3<br>(100%) | 0<br>(0%) | 13<br>(92.3%) |
|       | 1           | 2           | 3         | total         |

■ Accuracy ■ False ■ Precision ■ Size ■

1. 1st Place

Probability Heatmap

|     | 1   | 2  | 3  | Exp |
|-----|-----|----|----|-----|
| 2B  | 86  | 14 | 0  | 1   |
| 2C  | 85  | 15 | 0  | 1   |
| 2D  | 100 | 0  | 0  | 1   |
| 2E  | 83  | 17 | 0  | 1   |
| 2F  | 81  | 19 | 0  | 1   |
| 2I  | 92  | 8  | 0  | 1   |
| 2K  | 100 | 0  | 0  | 1   |
| 2N  | 77  | 23 | 0  | 1   |
| 2Q  | 98  | 2  | 0  | 1   |
| 2AD | 29  | 69 | 2  | 2   |
| 2AI | 5   | 78 | 17 | 2   |
| 2AL | 67  | 33 | 1  | 2   |
| 2AN | 10  | 81 | 8  | 2   |



1. 1st Place



## External Validation

### Confusion Matrix

|       |             |             |             |              |
|-------|-------------|-------------|-------------|--------------|
| 1     | 1<br>(100%) | 0<br>(0%)   | 0<br>(0%)   | 1<br>(16.7%) |
| 2     | 0<br>(0%)   | 3<br>(100%) | 0<br>(0%)   | 3<br>(50%)   |
| 3     | 0<br>(0%)   | 0<br>(0%)   | 2<br>(100%) | 2<br>(33.3%) |
| total | 1<br>(100%) | 3<br>(100%) | 2<br>(100%) | 6<br>(100%)  |
|       | 1           | 2           | 3           | total        |

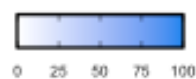
■ Accuracy ■ False ■ Precision ■ Size ■

1. 1st Place

[Back to scope – External Validation](#)

### Probability Heatmap

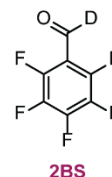
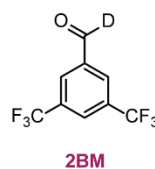
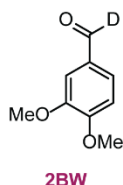
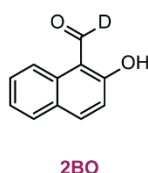
|     |    |    |     |     |
|-----|----|----|-----|-----|
|     | 1  | 2  | 3   | Exp |
| 2BD | 29 | 69 | 3   | 2   |
| 2BE | 5  | 78 | 17  | 2   |
| 2BF | 74 | 25 | 0   | 1   |
| 2BG | 3  | 69 | 28  | 2   |
| 2BH | 0  | 0  | 100 | 3   |
| 2BI | 1  | 34 | 66  | 3   |



1. 1st Place

## New Molecules - Prediction of Conditions

The following substrates were tested experimentally based on the model's predicted reaction condition. The table below presents the probability of each substrate belonging to each of the conditions categories.



|     | Predicted | 1   | 2  | 3   |
|-----|-----------|-----|----|-----|
| 2BM | 1         | 100 | 0  | 0   |
| 2BO | 3         | 0   | 0  | 100 |
| 2BS | 1         | 100 | 0  | 0   |
| 2BW | 2         | 1   | 58 | 41  |

[Back to scope – Prediction Set](#)

## Model Analysis - Variable Impact

To evaluate the impact of each of the features on the resulting classification model, we removed them from the model one feature at a time and analyzed the loss in overall accuracy as well as precision and recall for each category. In this case, omitting the dipole moment component most significantly decreased the training and validation data accuracies. The strong impact of this feature on the model likely originates from its ability to capture both steric and electronic contributions. In each feature removal, a with the corresponding formula is fitted to the same data, and its result as is produced by our code is presented.

To understand the analysis made in the SI it is recommended to not the recall (marked in green in the confusion table) and the accuracy (marked in blue in each confusion table) and compare it back to the original confusion table of the chosen model shown above. This comparison will illustrate the impact each feature has on the model.

## NPA\_1 Removed

formula

---

class ~ -2-3- + dip\_y + Dist(2, 7)

formula

---

#### Full Model Stats - Overall Accuracy and Pseudo-R2

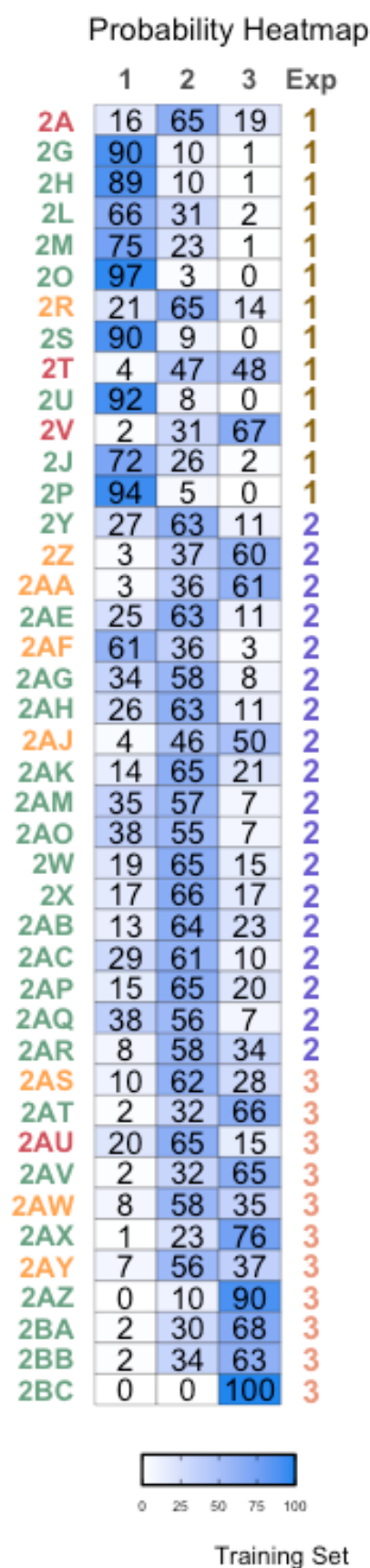
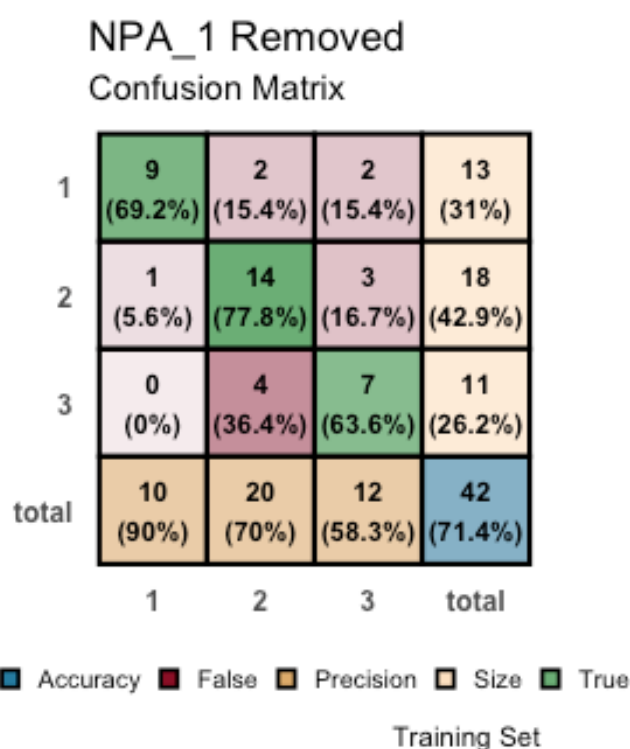
| Accuracy | McFadden_R2 |
|----------|-------------|
|----------|-------------|

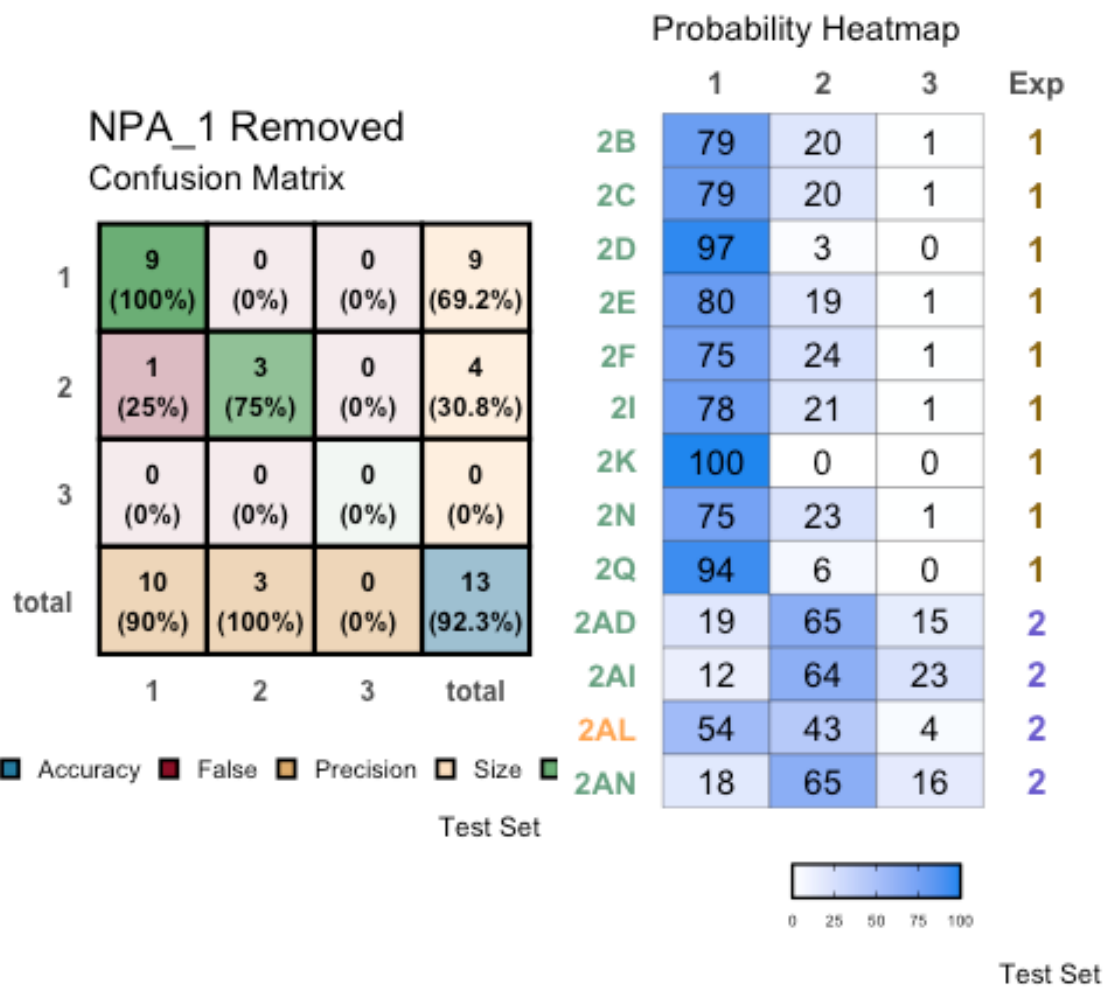
---

|        |       |
|--------|-------|
| 71.43% | 0.338 |
|--------|-------|

#### Model Coefficients

|            | x       |
|------------|---------|
| -2-3-      | -0.188  |
| dip_y      | -0.938  |
| Dist(2, 7) | 145.280 |





## NPA\_1 Removed

Confusion Matrix

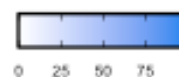
|       |             |             |             |              |
|-------|-------------|-------------|-------------|--------------|
| 1     | 1<br>(100%) | 0<br>(0%)   | 0<br>(0%)   | 1<br>(16.7%) |
| 2     | 0<br>(0%)   | 3<br>(100%) | 0<br>(0%)   | 3<br>(50%)   |
| 3     | 0<br>(0%)   | 0<br>(0%)   | 2<br>(100%) | 2<br>(33.3%) |
| total | 1<br>(100%) | 3<br>(100%) | 2<br>(100%) | 6<br>(100%)  |
|       | 1           | 2           | 3           | total        |

■ Accuracy ■ False ■ Precision ■ Size ■

External Validation

Probability Heatmap

|     |    |    |    |     |
|-----|----|----|----|-----|
|     | 1  | 2  | 3  | Exp |
| 2BD | 36 | 57 | 7  | 2   |
| 2BE | 5  | 48 | 48 | 2   |
| 2BF | 54 | 43 | 4  | 1   |
| 2BG | 8  | 58 | 34 | 2   |
| 2BH | 0  | 1  | 99 | 3   |
| 2BI | 4  | 44 | 52 | 3   |



External Validation

### Dist(2, 7) Removed

formula

---

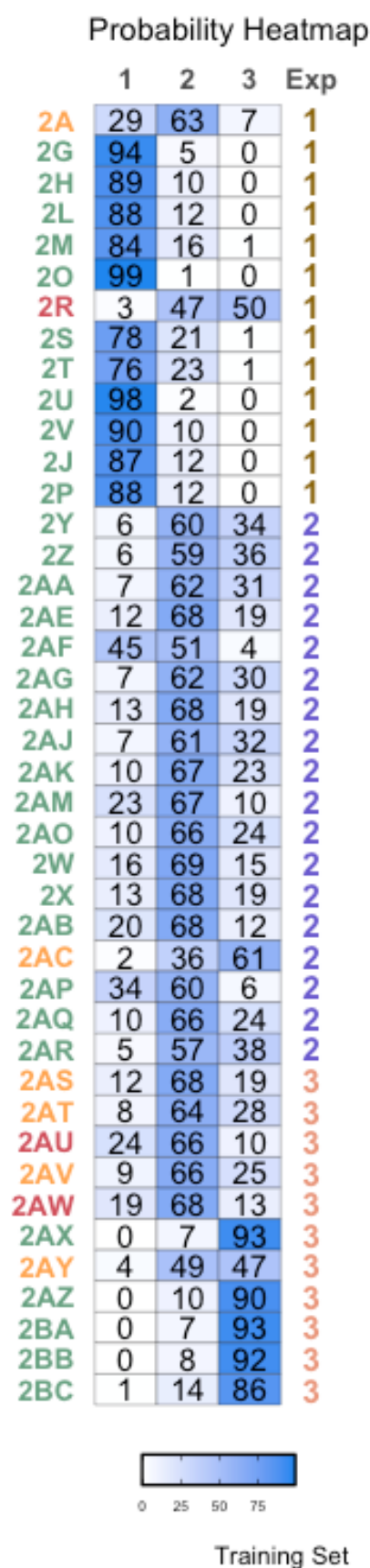
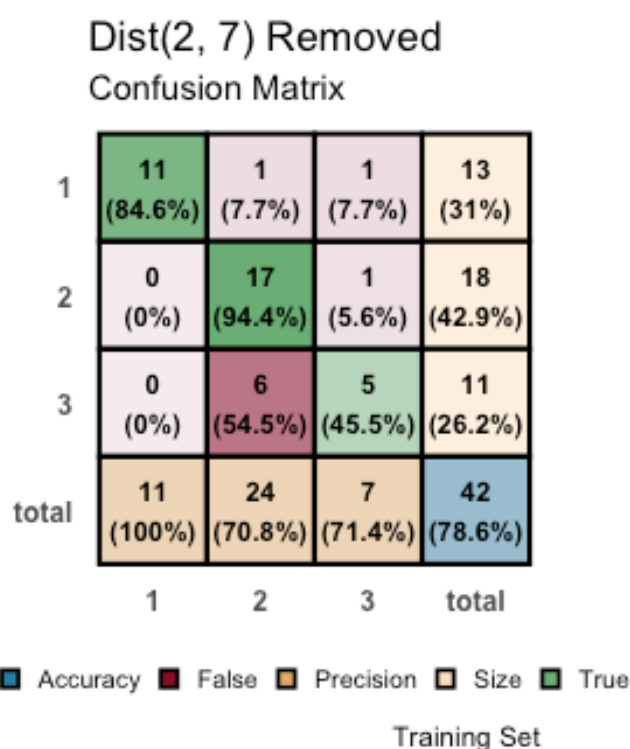
class ~ -2-3- + dip\_y + NPA\_1

Full Model Stats - Overall Accuracy and Pseudo-R2

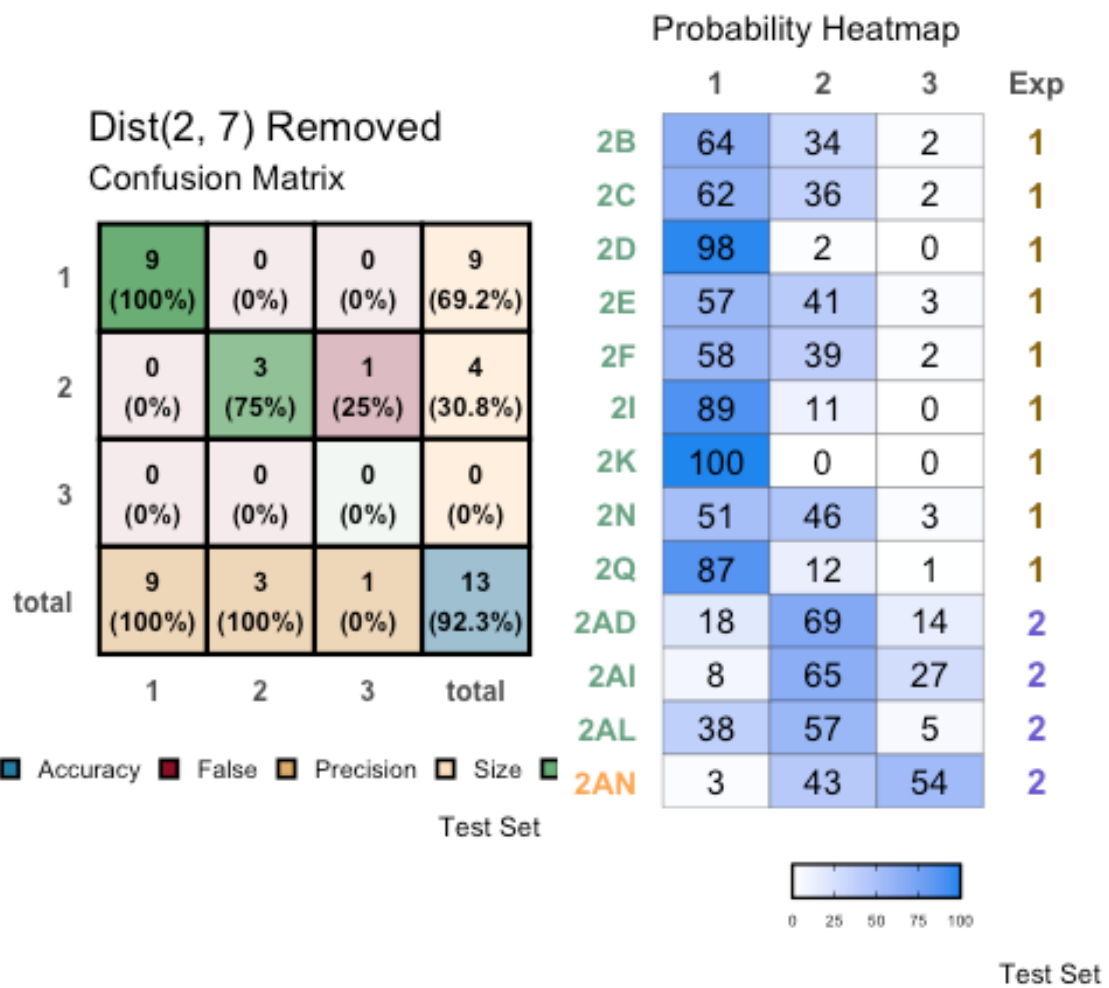
| Accuracy | McFadden_R2 |
|----------|-------------|
| 78.57%   | 0.452       |

Model Coefficients

|       | x      |
|-------|--------|
| -2-3- | -0.076 |
| dip_y | -1.830 |
| NPA_1 | 27.040 |







# Dist(2, 7) Removed Confusion Matrix

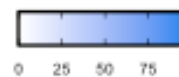
|       |             |             |             |              |
|-------|-------------|-------------|-------------|--------------|
| 1     | 1<br>(100%) | 0<br>(0%)   | 0<br>(0%)   | 1<br>(16.7%) |
| 2     | 0<br>(0%)   | 3<br>(100%) | 0<br>(0%)   | 3<br>(50%)   |
| 3     | 0<br>(0%)   | 0<br>(0%)   | 2<br>(100%) | 2<br>(33.3%) |
| total | 1<br>(100%) | 3<br>(100%) | 2<br>(100%) | 6<br>(100%)  |
|       | 1           | 2           | 3           | total        |

■ Accuracy
 ■ False
 ■ Precision
 ■ Size
 ■

External Validation

## Probability Heatmap

|     | 1  | 2  | 3  | Exp |
|-----|----|----|----|-----|
| 2BD | 24 | 66 | 10 | 2   |
| 2BE | 8  | 65 | 27 | 2   |
| 2BF | 68 | 31 | 2  | 1   |
| 2BG | 12 | 68 | 20 | 2   |
| 2BH | 0  | 3  | 97 | 3   |
| 2BI | 1  | 27 | 72 | 3   |



External Validation

### dip\_y Removed

formula

---

class ~ -2-3- + Dist(2, 7) + NPA\_1

Full Model Stats - Overall Accuracy and Pseudo-R2

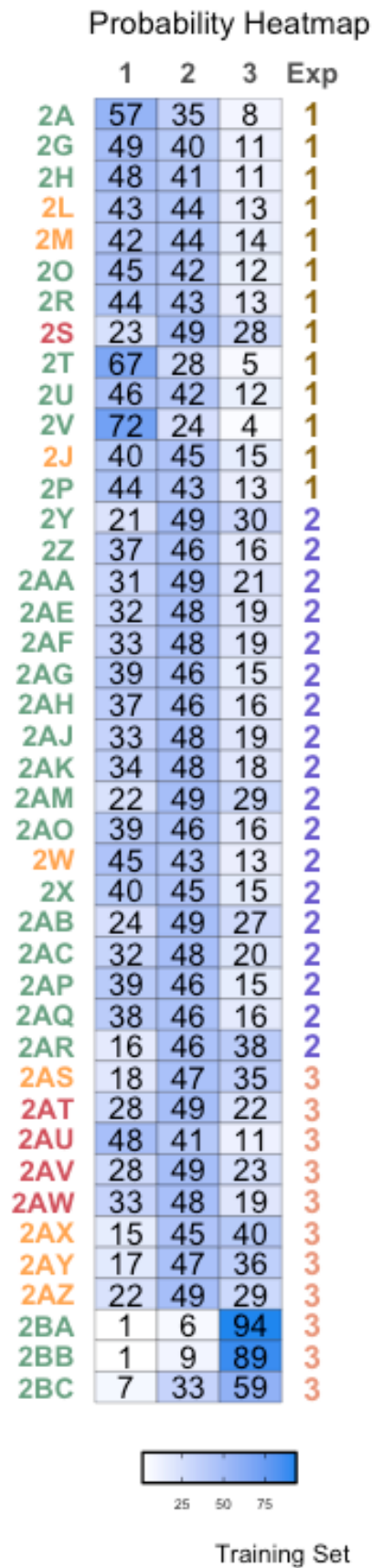
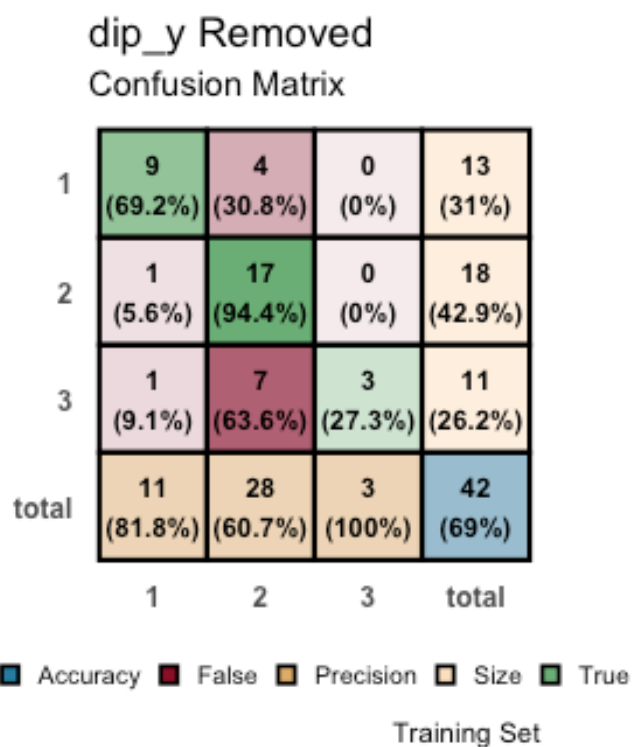
| Accuracy | McFadden_R2 |
|----------|-------------|
|----------|-------------|

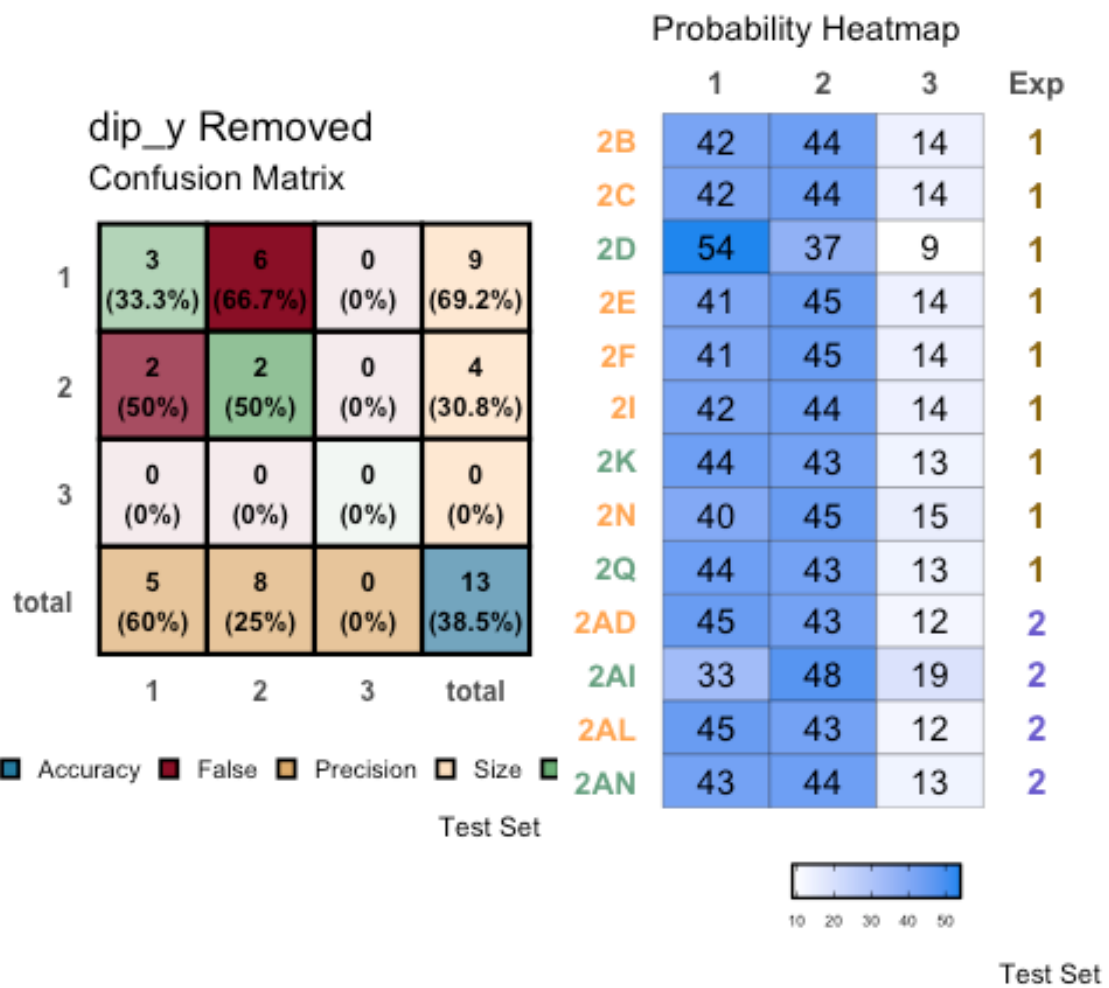
---

|        |       |
|--------|-------|
| 69.05% | 0.217 |
|--------|-------|

Model Coefficients

|            | x      |
|------------|--------|
| -2-3-      | -0.087 |
| Dist(2, 7) | 73.650 |
| NPA_1      | 15.303 |





dip\_y Removed  
Confusion Matrix

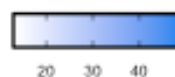
|       |           |             |           |              |
|-------|-----------|-------------|-----------|--------------|
| 1     | 0<br>(0%) | 1<br>(100%) | 0<br>(0%) | 1<br>(16.7%) |
| 2     | 0<br>(0%) | 3<br>(100%) | 0<br>(0%) | 3<br>(50%)   |
| 3     | 0<br>(0%) | 2<br>(100%) | 0<br>(0%) | 2<br>(33.3%) |
| total | 0<br>(0%) | 6<br>(50%)  | 0<br>(0%) | 6<br>(50%)   |
|       | 1         | 2           | 3         | total        |

■ Accuracy ■ False ■ Precision ■ Size ■

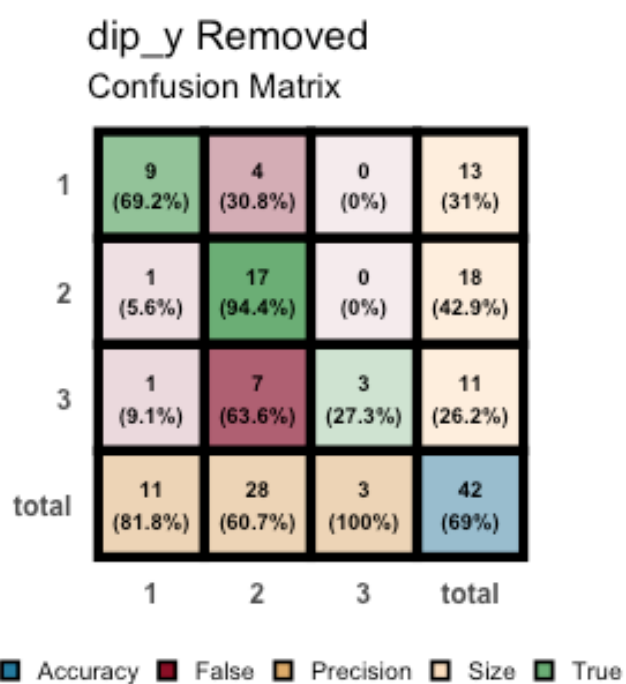
External Validation

Probability Heatmap

|     |    |    |    |     |
|-----|----|----|----|-----|
|     | 1  | 2  | 3  | Exp |
| 2BD | 37 | 46 | 16 | 2   |
| 2BE | 38 | 46 | 16 | 2   |
| 2BF | 43 | 44 | 13 | 1   |
| 2BG | 29 | 49 | 22 | 2   |
| 2BH | 18 | 48 | 34 | 3   |
| 2BI | 29 | 49 | 22 | 3   |



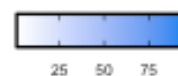
External Validation



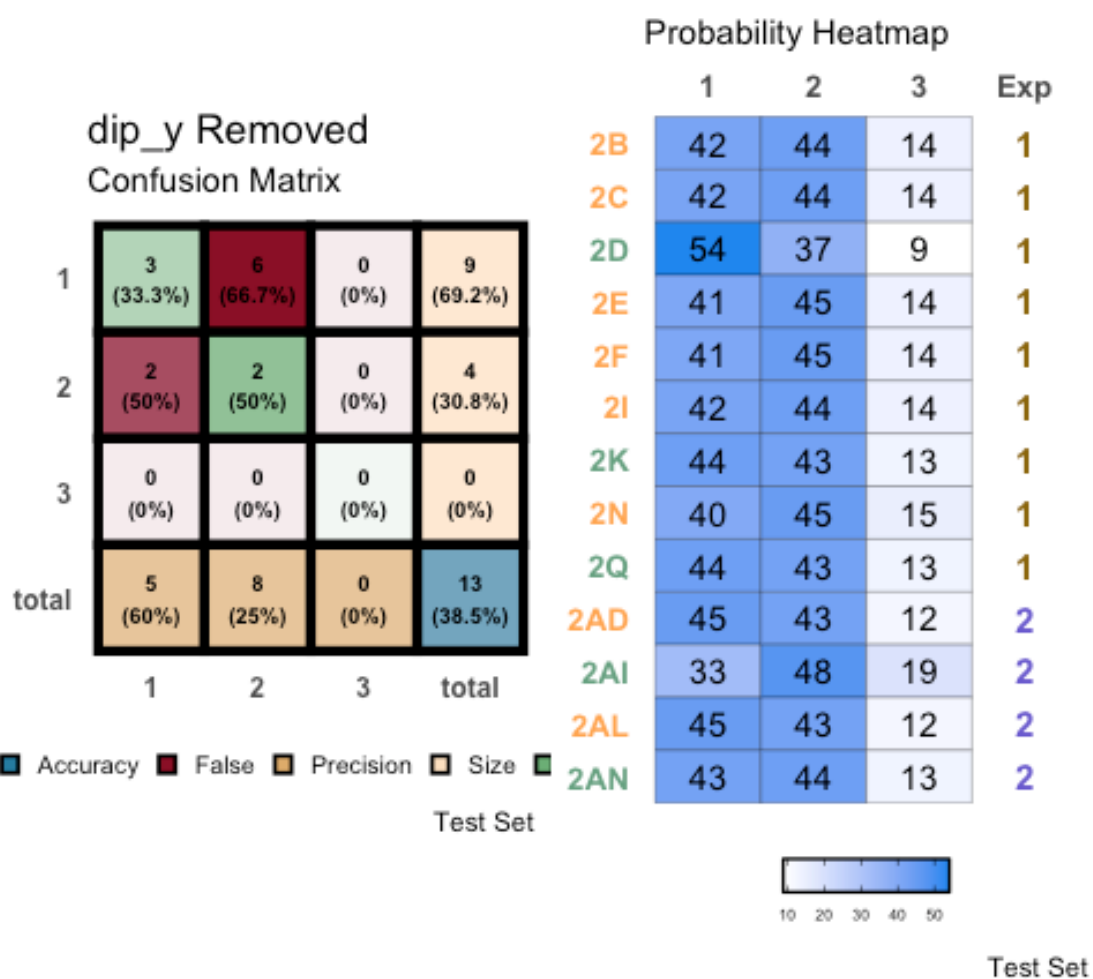
Training Set

Probability Heatmap

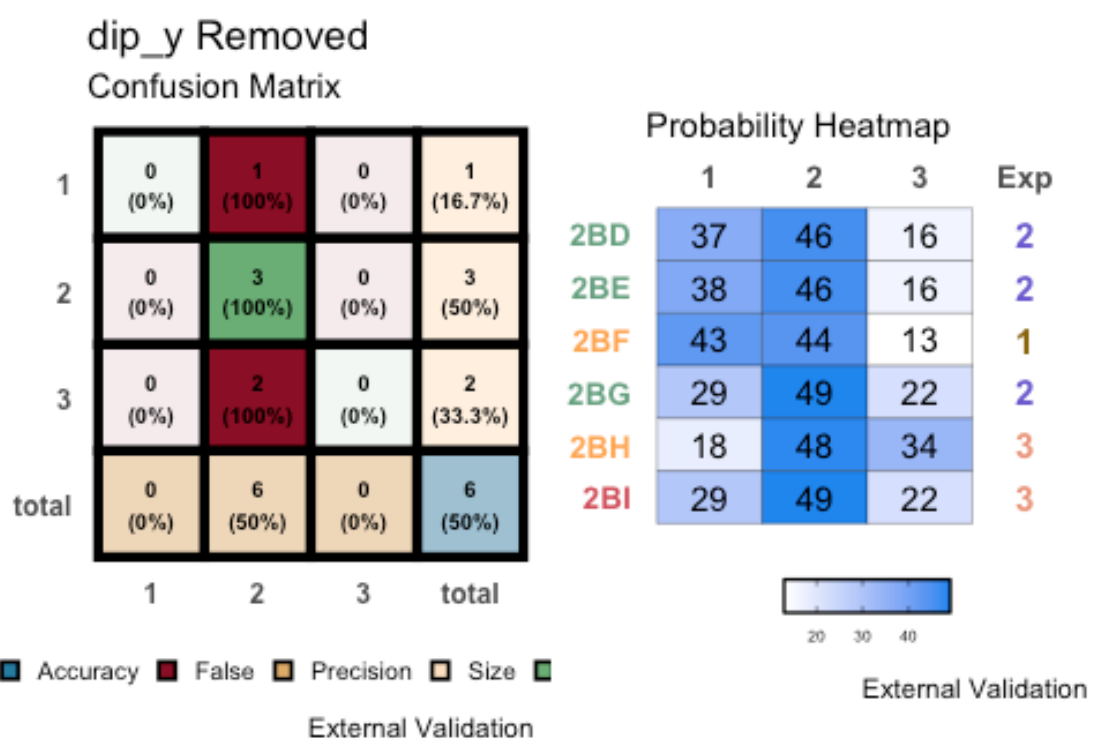
|     | 1  | 2  | 3  | Exp |
|-----|----|----|----|-----|
| 2A  | 57 | 35 | 8  | 1   |
| 2G  | 49 | 40 | 11 | 1   |
| 2H  | 48 | 41 | 11 | 1   |
| 2L  | 43 | 44 | 13 | 1   |
| 2M  | 42 | 44 | 14 | 1   |
| 2O  | 45 | 42 | 12 | 1   |
| 2R  | 44 | 43 | 13 | 1   |
| 2S  | 23 | 49 | 28 | 1   |
| 2T  | 67 | 28 | 5  | 1   |
| 2U  | 46 | 42 | 12 | 1   |
| 2V  | 72 | 24 | 4  | 1   |
| 2J  | 40 | 45 | 15 | 1   |
| 2P  | 44 | 43 | 13 | 1   |
| 2Y  | 21 | 49 | 30 | 2   |
| 2Z  | 37 | 46 | 16 | 2   |
| 2AA | 31 | 49 | 21 | 2   |
| 2AE | 32 | 48 | 19 | 2   |
| 2AF | 33 | 48 | 19 | 2   |
| 2AG | 39 | 46 | 15 | 2   |
| 2AH | 37 | 46 | 16 | 2   |
| 2AJ | 33 | 48 | 19 | 2   |
| 2AK | 34 | 48 | 18 | 2   |
| 2AM | 22 | 49 | 29 | 2   |
| 2AO | 39 | 46 | 16 | 2   |
| 2W  | 45 | 43 | 13 | 2   |
| 2X  | 40 | 45 | 15 | 2   |
| 2AB | 24 | 49 | 27 | 2   |
| 2AC | 32 | 48 | 20 | 2   |
| 2AP | 39 | 46 | 15 | 2   |
| 2AQ | 38 | 46 | 16 | 2   |
| 2AR | 16 | 46 | 38 | 2   |
| 2AS | 18 | 47 | 35 | 3   |
| 2AT | 28 | 49 | 22 | 3   |
| 2AU | 48 | 41 | 11 | 3   |
| 2AV | 28 | 49 | 23 | 3   |
| 2AW | 33 | 48 | 19 | 3   |
| 2AX | 15 | 45 | 40 | 3   |
| 2AY | 17 | 47 | 36 | 3   |
| 2AZ | 22 | 49 | 29 | 3   |
| 2BA | 1  | 6  | 94 | 3   |
| 2BB | 1  | 9  | 89 | 3   |
| 2BC | 7  | 33 | 59 | 3   |



Training Set







### **-2-3- Removed**

formula

---

class ~ dip\_y + Dist(2, 7) + NPA\_1

Full Model Stats - Overall Accuracy and Pseudo-R2

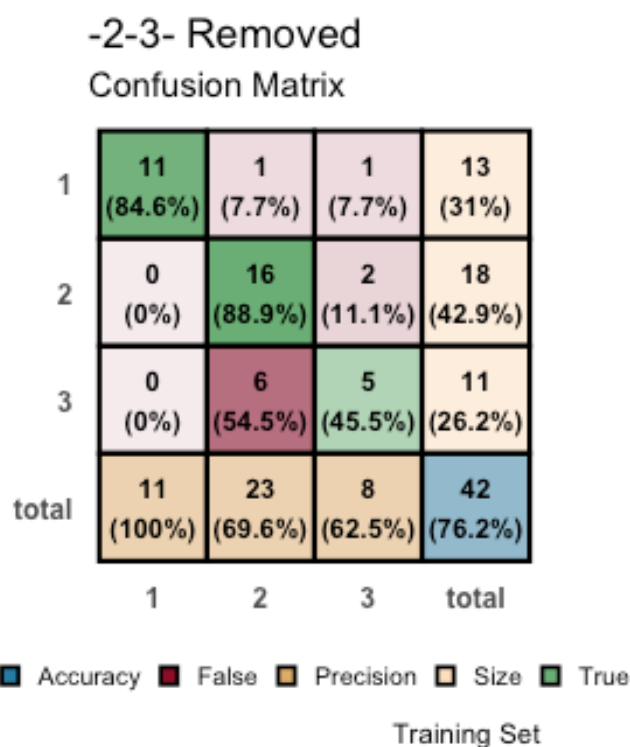
| Accuracy | McFadden_R2 |
|----------|-------------|
|----------|-------------|

---

|        |       |
|--------|-------|
| 76.19% | 0.344 |
|--------|-------|

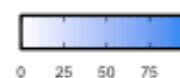
Model Coefficients

|            | x      |
|------------|--------|
| dip_y      | -1.932 |
| Dist(2, 7) | -7.774 |
| NPA_1      | 23.524 |

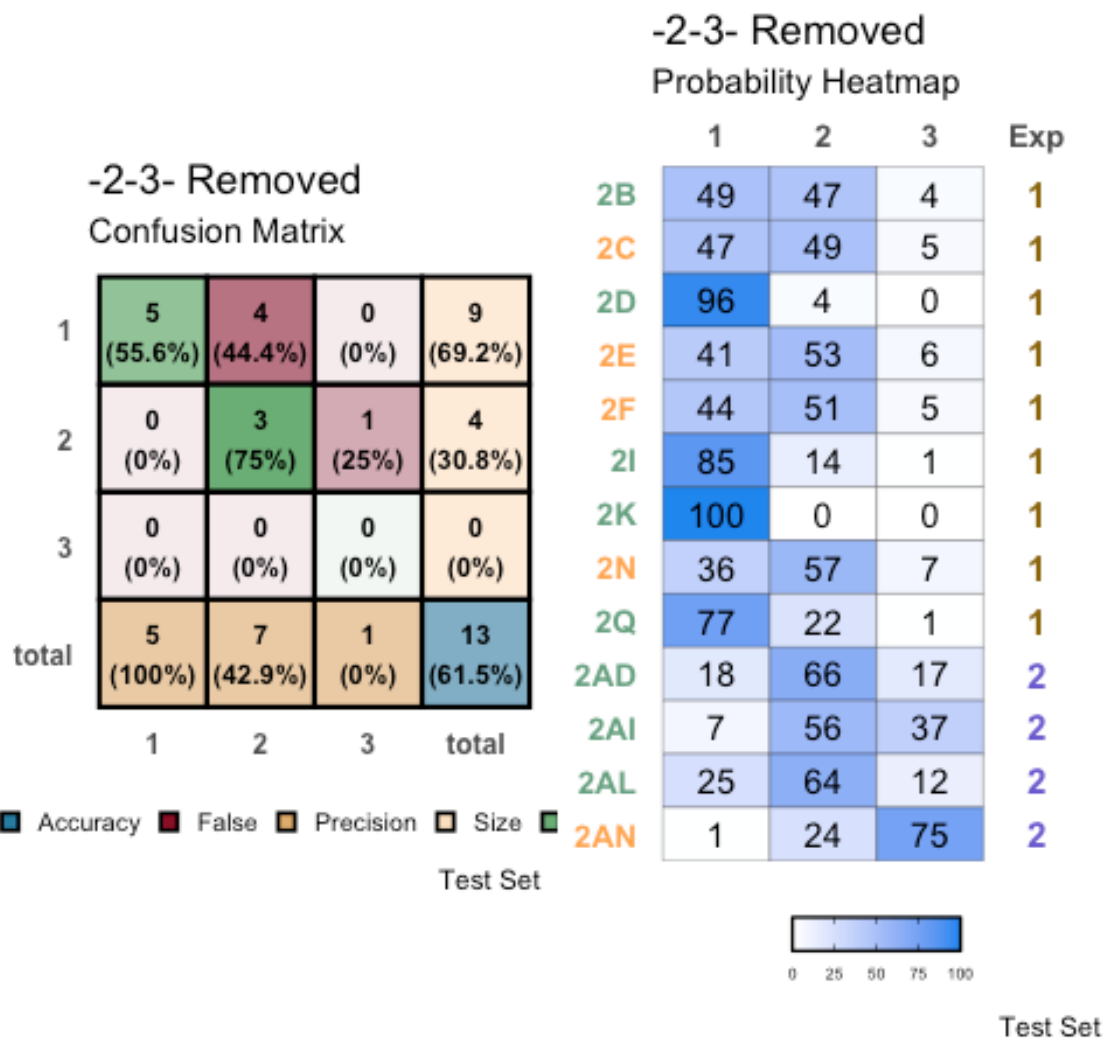


-2-3- Removed  
Probability Heatmap

|     | 1  | 2  | 3  | Exp |
|-----|----|----|----|-----|
| 2A  | 27 | 63 | 11 | 1   |
| 2G  | 91 | 9  | 0  | 1   |
| 2H  | 81 | 18 | 1  | 1   |
| 2L  | 85 | 15 | 1  | 1   |
| 2M  | 78 | 21 | 1  | 1   |
| 2O  | 98 | 2  | 0  | 1   |
| 2R  | 2  | 26 | 72 | 1   |
| 2S  | 80 | 19 | 1  | 1   |
| 2T  | 77 | 22 | 1  | 1   |
| 2U  | 97 | 3  | 0  | 1   |
| 2V  | 92 | 8  | 0  | 1   |
| 2J  | 84 | 15 | 1  | 1   |
| 2P  | 78 | 21 | 1  | 1   |
| 2Y  | 7  | 57 | 36 | 2   |
| 2Z  | 8  | 59 | 32 | 2   |
| 2AA | 12 | 64 | 24 | 2   |
| 2AE | 9  | 61 | 30 | 2   |
| 2AF | 35 | 57 | 7  | 2   |
| 2AG | 4  | 44 | 52 | 2   |
| 2AH | 9  | 60 | 32 | 2   |
| 2AJ | 7  | 56 | 38 | 2   |
| 2AK | 8  | 59 | 33 | 2   |
| 2AM | 30 | 61 | 9  | 2   |
| 2AO | 5  | 52 | 43 | 2   |
| 2W  | 16 | 66 | 18 | 2   |
| 2X  | 14 | 65 | 22 | 2   |
| 2AB | 28 | 62 | 10 | 2   |
| 2AC | 2  | 28 | 71 | 2   |
| 2AP | 37 | 56 | 7  | 2   |
| 2AQ | 5  | 52 | 43 | 2   |
| 2AR | 10 | 62 | 28 | 2   |
| 2AS | 22 | 65 | 13 | 3   |
| 2AT | 16 | 65 | 19 | 3   |
| 2AU | 24 | 64 | 12 | 3   |
| 2AV | 18 | 66 | 17 | 3   |
| 2AW | 20 | 65 | 15 | 3   |
| 2AX | 0  | 8  | 92 | 3   |
| 2AY | 7  | 56 | 37 | 3   |
| 2AZ | 0  | 10 | 90 | 3   |
| 2BA | 3  | 37 | 61 | 3   |
| 2BB | 1  | 19 | 80 | 3   |
| 2BC | 3  | 40 | 56 | 3   |



Training Set



-2-3- Removed  
Confusion Matrix

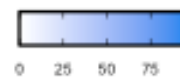
|       |             |             |             |              |
|-------|-------------|-------------|-------------|--------------|
| 1     | 1<br>(100%) | 0<br>(0%)   | 0<br>(0%)   | 1<br>(16.7%) |
| 2     | 0<br>(0%)   | 3<br>(100%) | 0<br>(0%)   | 3<br>(50%)   |
| 3     | 0<br>(0%)   | 0<br>(0%)   | 2<br>(100%) | 2<br>(33.3%) |
| total | 1<br>(100%) | 3<br>(100%) | 2<br>(100%) | 6<br>(100%)  |
|       | 1           | 2           | 3           | total        |

■ Accuracy ■ False ■ Precision ■ Size ■

External Validation

-2-3- Removed  
Probability Heatmap

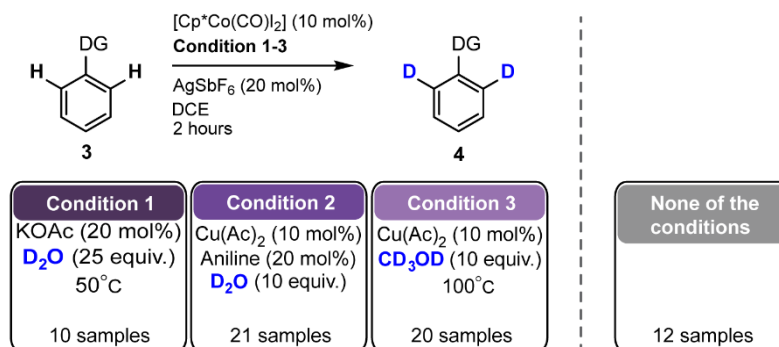
|     |    |    |    |     |
|-----|----|----|----|-----|
|     | 1  | 2  | 3  | Exp |
| 2BD | 18 | 66 | 17 | 2   |
| 2BE | 20 | 65 | 14 | 2   |
| 2BF | 59 | 38 | 3  | 1   |
| 2BG | 12 | 64 | 24 | 2   |
| 2BH | 0  | 4  | 96 | 3   |
| 2BI | 1  | 20 | 79 | 3   |



External Validation

## 2.2 Cobalt catalyzed hydrogen isotope exchange (HIE)

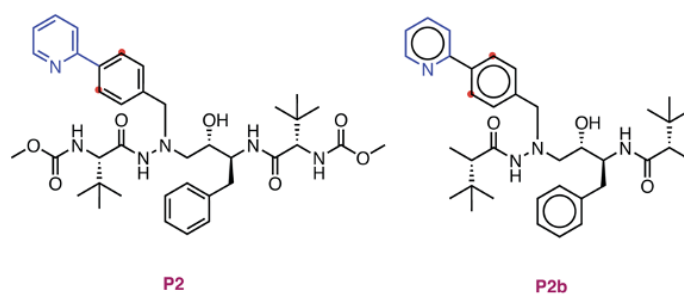
The following sections contain the list of best models as extracted from the code in R provided in our GitHub account. The scopes and reaction details are given in the main SI. The reaction to which the following model attributes is as follows:



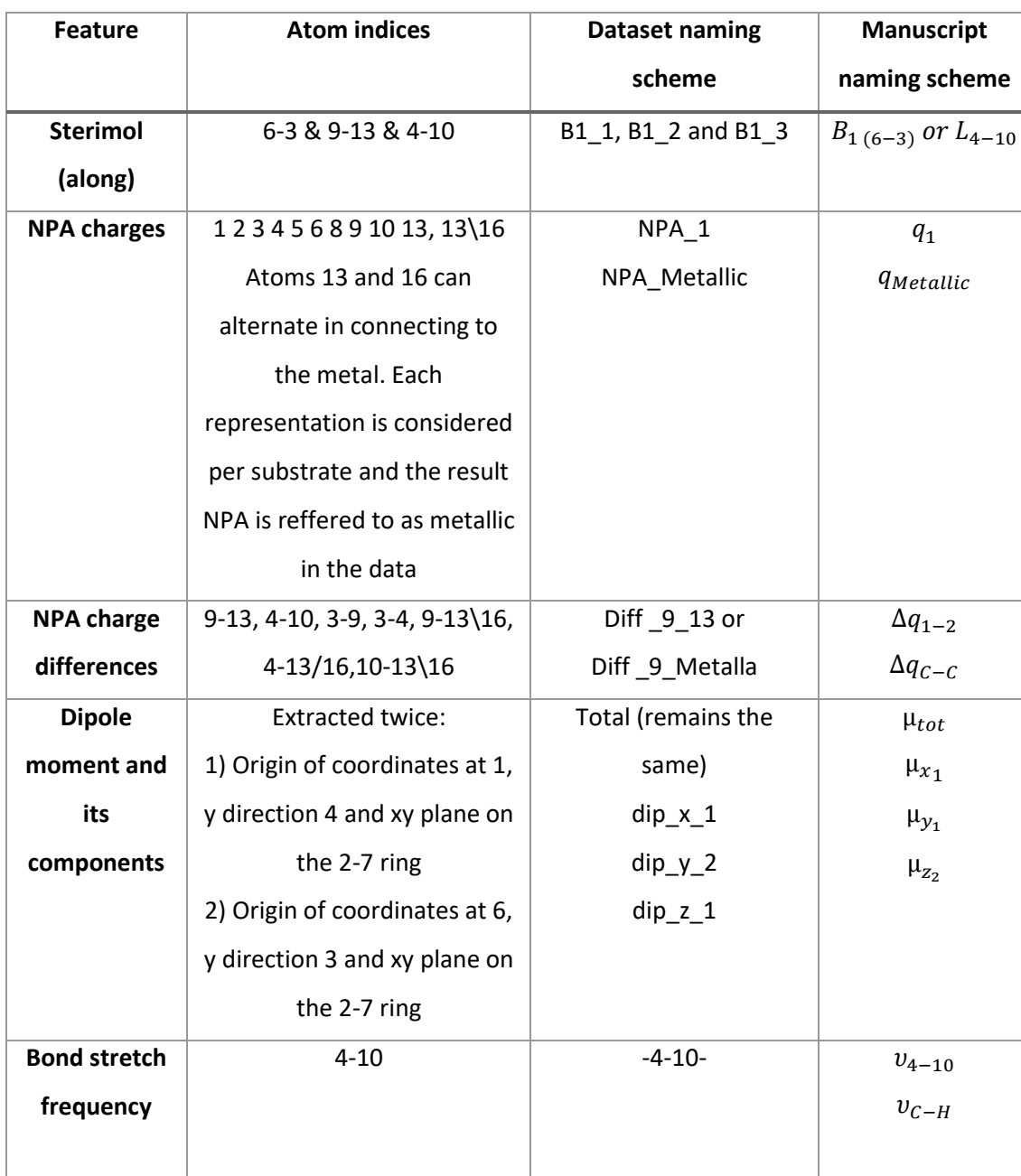
**Figure S3:** Representation of the reaction

### 2.2.1 Features extracted for model identification

Information about the different RDKit fragments, their names and definitions are available on the modules [webpage](#). We screened out features that had no variance (all the same value). In this case study one of the molecules in the prediction set had a value of 4 amides, where the maximum value in the trained model was 2. We decided to cap the number of amides to the maximal number in the training set. We tested setting its value to 0, 1, and 2 (see figure S4) and its predicted conditions was not affected. We presume this stems from the location of the amides in this substrate. The other molecules in the set that have amides, have them in functional positions, whereas in this molecule they are far from the reaction site.



**Figure S4:** capping the number of amides fragments based on the training set



|                                                |                                                          |                    |                                    |
|------------------------------------------------|----------------------------------------------------------|--------------------|------------------------------------|
| <b>Bond lengths and distance between atoms</b> | 9-13, 3-9, 3-4, 4-10,                                    | Dist(9, 13)        | $d_{9-13}$<br>$d_{9-Metallic}$     |
| <b>Angles (dihedrals and between bonds)</b>    | D(9 3 4 10)<br>D(4 3 9 13)<br>Ang(4 3 13)<br>Ang(4 3 16) | Dihedral(9_3_4_10) | $dih_{9,3,4,10}$<br>$ang_{4,3,13}$ |

**Table S2:** Representation of the atom indexing and feature set

### 2.2.2 Best model and model identification reports

Below is the data for the model presented in the main text as it appears in the output generated by our statistical code.

The top 10 models and their analyses as they resulted from the model identification code can be found on the project page on GitHub – [Top 10 models CH Activation – RDKit Fragments](#)

The top 10 models using the stereoelectronic feature set also be found on the project page on GitHub – [Top 10 models CH Activation – Stereoelectronic](#)

The fragments that appeared in the most predictive model were amides, aldehydes, pyridines, and tertiary amines. Each of these fragments appear predominantly within a certain category; however, they are still spread across all categories. Nevertheless, the model can prioritize these fragments given a molecule containing two functional groups that may point to different conditions. For example, most of the aldehydes appear under **Condition 2** and most of the pyridines appear under **Condition 1**, however substrate **3E**, which contains both a pyridine and an aldehyde is correctly classified under **Condition 1** (see section 3.5.4, for another similar example substrate **3Z** in **Condition 3**).

These are the top-10 fitted models, as they result from the brute-force model search. In this case, the first-ranked model provided the best testing result and was, therefore, further analyzed.

Below are the confusion matrices and probability heatmaps for the training, test, external validation, and test sets, as they result from our code. Next to each of them, there is a link that takes you back to the scope, with which you should be able to move back and forth.

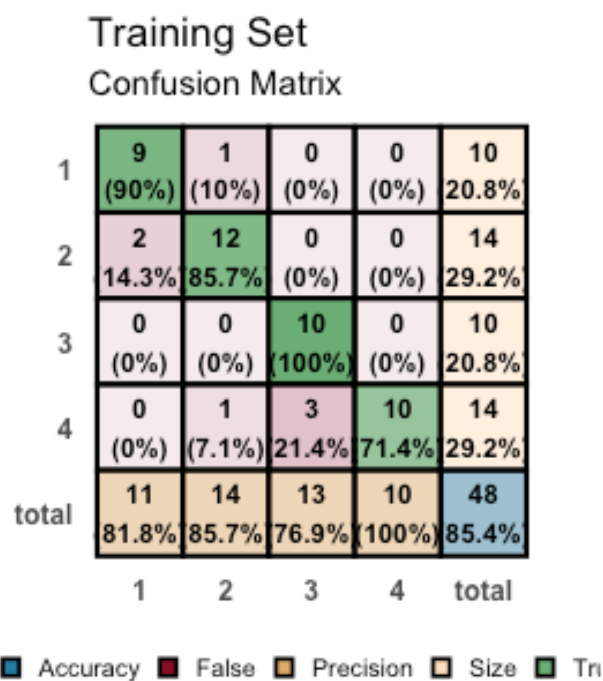


| formula                                                                 | McFadden<br>R2 |
|-------------------------------------------------------------------------|----------------|
| class ~ fr_NH0 + fr_aldehyde + fr_amide + fr_para_hydroxylation         | 0.766          |
| class ~ fr_Ar_N + fr_aldehyde + fr_amide + fr_para_hydroxylation        | 0.764          |
| class ~ fr_NH0 + fr_aldehyde + fr_amide + fr_pyridine                   | 0.762          |
| class ~ fr_aldehyde + fr_amide + fr_para_hydroxylation +<br>fr_pyridine | 0.755          |
| class ~ fr_Ar_N + fr_NH2 + fr_aldehyde + fr_amide                       | 0.738          |
| class ~ fr_C_O + fr_NH0 + fr_NH1 + fr_aldehyde                          | 0.737          |
| class ~ fr_C_O_noCOO + fr_NH0 + fr_NH1 + fr_aldehyde                    | 0.737          |
| class ~ fr_NH0 + fr_NH1 + fr_aldehyde + fr_amide                        | 0.737          |
| class ~ fr_NH0 + fr_NH1 + fr_aldehyde + fr_pyridine                     | 0.737          |
| class ~ fr_NH0 + fr_NH2 + fr_aldehyde + fr_amide                        | 0.737          |

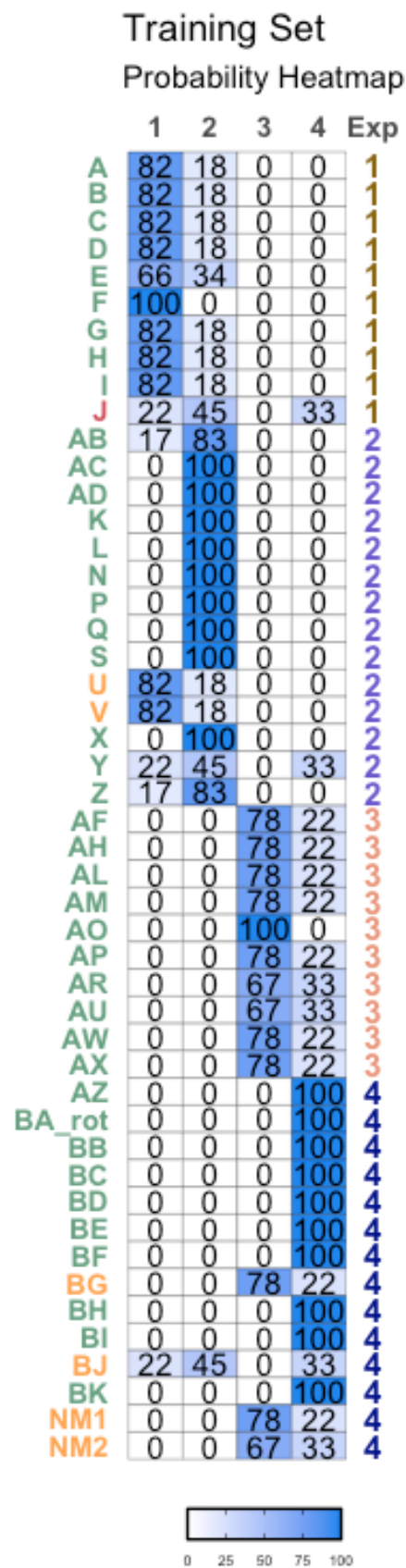
### Published Model

|                                                       |             |
|-------------------------------------------------------|-------------|
| formula                                               |             |
| class ~ fr_NH0 + fr_aldehyde + fr_amide + fr_pyridine |             |
| Full Model Stats - Overall Accuracy and Pseudo-R2     |             |
| Accuracy                                              | McFadden_R2 |
| 85.42%                                                | 0.762       |

[Back to scope – Training Set](#)



3. 3rd Place



3. 3rd Place

[Back to scope – Test Set](#)

Test Set

Confusion Matrix

|       |           |              |              |           |               |
|-------|-----------|--------------|--------------|-----------|---------------|
| 1     | 0<br>(0%) | 0<br>(0%)    | 0<br>(0%)    | 0<br>(0%) | 0<br>(0%)     |
| 2     | 0<br>(0%) | 4<br>(100%)  | 0<br>(0%)    | 0<br>(0%) | 4<br>(33.3%)  |
| 3     | 0<br>(0%) | 1<br>(12.5%) | 7<br>(87.5%) | 0<br>(0%) | 8<br>(66.7%)  |
| 4     | 0<br>(0%) | 0<br>(0%)    | 0<br>(0%)    | 0<br>(0%) | 0<br>(0%)     |
| total | 0<br>(0%) | 5<br>(80%)   | 7<br>(100%)  | 0<br>(0%) | 12<br>(91.7%) |
|       | 1         | 2            | 3            | 4         | total         |

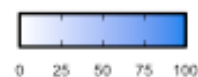
■ Accuracy ■ False ■ Precision ■ Size ■

3. 3rd Place

Test Set

Probability Heatmap

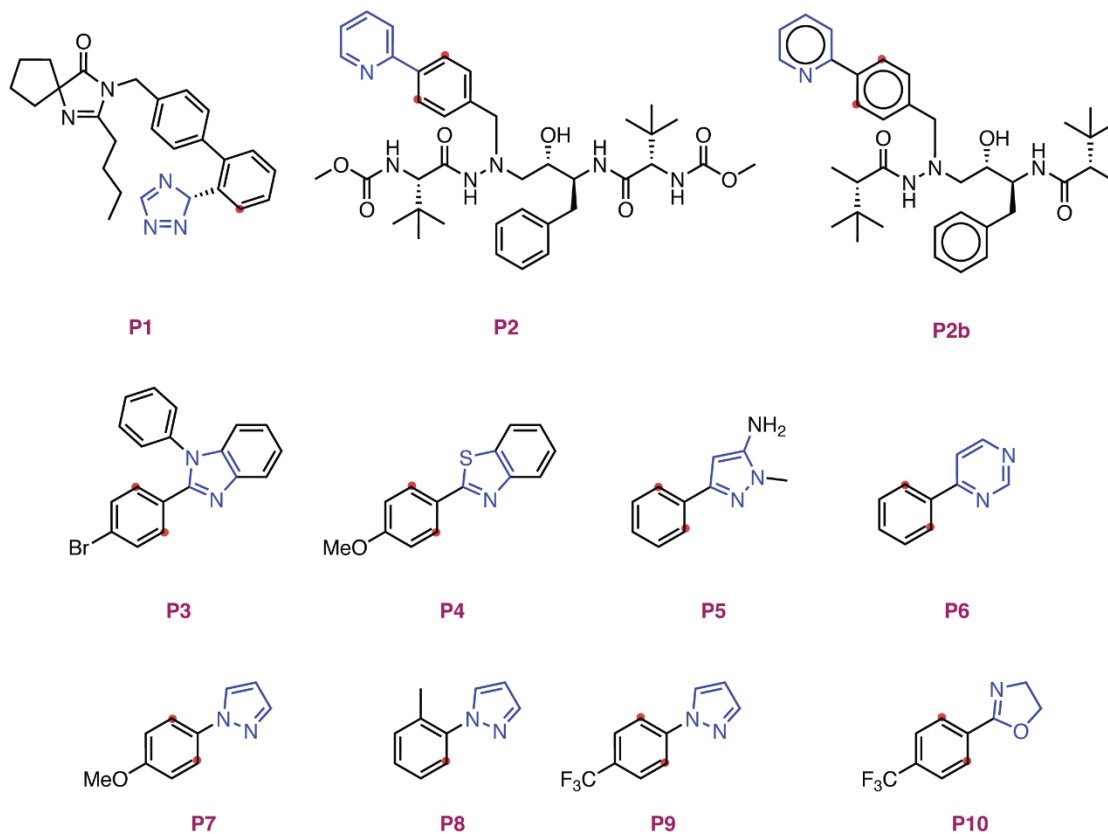
|    |    |     |    |    |     |
|----|----|-----|----|----|-----|
|    | 1  | 2   | 3  | 4  | Exp |
| M  | 0  | 100 | 0  | 0  | 2   |
| O  | 0  | 100 | 0  | 0  | 2   |
| W  | 0  | 100 | 0  | 0  | 2   |
| AK | 0  | 0   | 78 | 22 | 3   |
| AN | 0  | 0   | 78 | 22 | 3   |
| AV | 0  | 0   | 67 | 33 | 3   |
| T  | 0  | 100 | 0  | 0  | 2   |
| AT | 0  | 0   | 67 | 33 | 3   |
| AS | 0  | 0   | 67 | 33 | 3   |
| AQ | 22 | 45  | 0  | 33 | 3   |
| AI | 0  | 0   | 78 | 22 | 3   |
| AG | 0  | 0   | 78 | 22 | 3   |



3. 3rd Place

## New Molecules - Prediction of Conditions

The following substrates were tested experimentally based on the model's predicted reaction condition. The table below presents the probability of each substrate belonging to each of the conditions categories.



[Back to scope – Predictions Set](#)

|    | Prediction | 1   | 2   | 3 | 4   |
|----|------------|-----|-----|---|-----|
| p1 | 4          | 0   | 0   | 0 | 100 |
| p2 | 1          | 100 | 0   | 0 | 0   |
| p3 | 2          | 22  | 45  | 0 | 33  |
| p4 | 2          | 0   | 100 | 0 | 0   |
| p5 | 2          | 22  | 45  | 0 | 33  |
| p6 | 2          | 22  | 45  | 0 | 33  |
| p7 | 2          | 22  | 45  | 0 | 33  |
| p8 | 2          | 22  | 45  | 0 | 33  |
| p9 | 2          | 22  | 45  | 0 | 33  |

|     | Prediction | 1 | 2   | 3 | 4 |
|-----|------------|---|-----|---|---|
| p10 | 2          | 0 | 100 | 0 | 0 |

### Model Analysis - Variable Impact

To evaluate the impact of the fragment features on the resulting classification model, we removed each from the model and analyzed the loss in accuracy similarly to the previous case study (section 3.5.4). In this case, omitting the amide and aldehyde fragments most significantly decreased the overall training and validation data accuracy. Omitting each of these fragments decreased the model's capacity to classify these two directing groups that predominantly belong to **Conditions 3** and **2**, respectively. In addition, the omission of the amide fragment impacted the distinction between **Conditions 2, 3** and **4**, whereas the omission of the aldehyde fragment mainly impacted the distinction between **Conditions 2** and **4**. The omission of the pyridine, which was less impactful on the overall accuracy, impacted the distinction between **Conditions 1** and **2**. Finally, the omission of the tertiary amine, which was least impactful on the overall accuracy, affected the distinction between **Conditions 2** and **4**. This analysis further stresses the point that the fragments do not each define one of the conditions, but rather serve to collectively create nuanced distinctions that are not easily discerned intuitively. These are the direct results from our code, which generates these plots and summaries.

To understand the analysis made in the SI it is recommended to not the recall (marked in green in the confusion table) and the accuracy (marked in blue in each confusion table) and compare it back to the original confusion table of the chosen model shown above. This comparison will illustrate the impact each feature has on the model.

#### fr\_NH0 Removed

formula

---

class ~ fr\_aldehyde + fr\_amide + fr\_pyridine

Full Model Stats - Overall Accuracy and Pseudo-R2

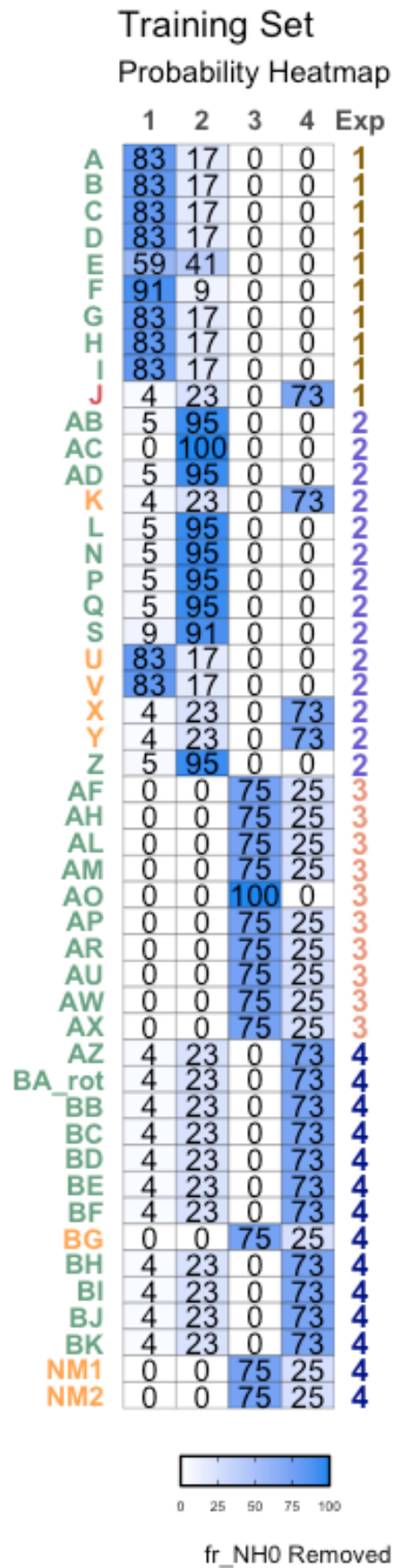
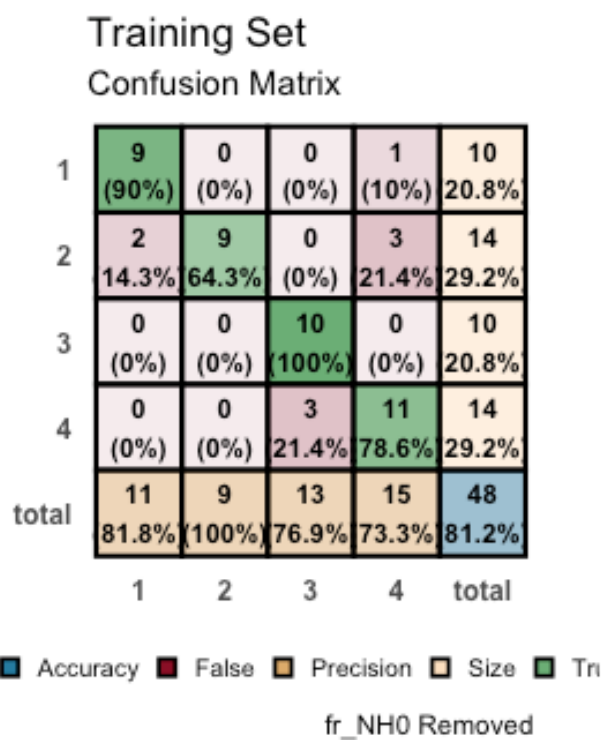
Accuracy    McFadden\_R2

---

81.25%    0.639

# Model Coefficients

|   | (Intercept) | fr_aldehyde1 | fr_amide1 | fr_amide2 | fr_pyridine1 | fr_pyridine2 |
|---|-------------|--------------|-----------|-----------|--------------|--------------|
| 2 | 1.761       | 1.263        | -0.705    | -14.793   | -3.373       | 23.695       |
| 3 | -16.420     | -56.180      | 51.236    | 47.976    | -59.021      | -1.446       |
| 4 | 2.931       | -65.434      | 30.786    | -21.319   | -46.467      | -15.186      |



Test Set

Confusion Matrix

|       |           |             |             |            |             |
|-------|-----------|-------------|-------------|------------|-------------|
| 1     | 0<br>(0%) | 0<br>(0%)   | 0<br>(0%)   | 0<br>(0%)  | 0<br>(0%)   |
| 2     | 0<br>(0%) | 2<br>(50%)  | 0<br>(0%)   | 2<br>(50%) | 4<br>33.3%  |
| 3     | 0<br>(0%) | 0<br>(0%)   | 7<br>87.5%  | 1<br>12.5% | 8<br>66.7%  |
| 4     | 0<br>(0%) | 0<br>(0%)   | 0<br>(0%)   | 0<br>(0%)  | 0<br>(0%)   |
| total | 0<br>(0%) | 2<br>(100%) | 7<br>(100%) | 3<br>(0%)  | 12<br>(75%) |
|       | 1         | 2           | 3           | 4          | total       |

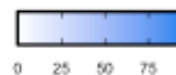
■ Accuracy ■ False ■ Precision ■ Size ■

fr\_NH0 Removed

## Test Set

### Probability Heatmap

|    | 1 | 2  | 3  | 4  | Exp |
|----|---|----|----|----|-----|
| M  | 5 | 95 | 0  | 0  | 2   |
| O  | 5 | 95 | 0  | 0  | 2   |
| W  | 4 | 23 | 0  | 73 | 2   |
| AK | 0 | 0  | 75 | 25 | 3   |
| AN | 0 | 0  | 75 | 25 | 3   |
| AV | 0 | 0  | 75 | 25 | 3   |
| T  | 4 | 23 | 0  | 73 | 2   |
| AT | 0 | 0  | 75 | 25 | 3   |
| AS | 0 | 0  | 75 | 25 | 3   |
| AQ | 4 | 23 | 0  | 73 | 3   |
| AI | 0 | 0  | 75 | 25 | 3   |
| AG | 0 | 0  | 75 | 25 | 3   |



fr\_NH0 Removed



### fr\_aldehyde Removed

formula

---

class ~ fr\_NH0 + fr\_amide + fr\_pyridine

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy    McFadden\_R2

---

72.92%    0.54

### Model Coefficients

|   | (Intercept<br>) | fr_NH0<br>1 | fr_NH0<br>2 | fr_NH0<br>5 | fr_amide<br>1 | fr_amide<br>2 | fr_pyridine<br>1 | fr_pyridine<br>2 |
|---|-----------------|-------------|-------------|-------------|---------------|---------------|------------------|------------------|
| 2 | 34.562          | -5.124      | -33.463     | -13.301     | -14.270       | -18.966       | -30.823          | 22.846           |
| 3 | -4.246          | -6.345      | -15.303     | -1.365      | 27.161        | 47.312        | -45.927          | -1.138           |
| 4 | 35.000          | -6.682      | -35.000     | 18.090      | -13.118       | -21.087       | -52.781          | -10.537          |

Training Set  
Confusion Matrix

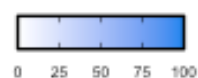
|       |             |             |             |             |             |
|-------|-------------|-------------|-------------|-------------|-------------|
| 1     | 9<br>(90%)  | 1<br>(10%)  | 0<br>(0%)   | 0<br>(0%)   | 10<br>20.8% |
| 2     | 2<br>14.3%  | 6<br>42.9%  | 1<br>(7.1%) | 5<br>35.7%  | 14<br>29.2% |
| 3     | 0<br>(0%)   | 0<br>(0%)   | 10<br>100%  | 0<br>(0%)   | 10<br>20.8% |
| 4     | 0<br>(0%)   | 1<br>(7.1%) | 3<br>21.4%  | 10<br>71.4% | 14<br>29.2% |
| total | 11<br>81.8% | 8<br>(75%)  | 14<br>71.4% | 15<br>66.7% | 48<br>72.9% |
|       | 1           | 2           | 3           | 4           | total       |

■ Accuracy ■ False ■ Precision ■ Size ■ Tr

fr\_aldehyde Removed

Training Set  
Probability Heatmap

|        |     |     |     |     |     |
|--------|-----|-----|-----|-----|-----|
|        | 1   | 2   | 3   | 4   | Exp |
| A      | 80  | 20  | 0   | 0   | 1   |
| B      | 80  | 20  | 0   | 0   | 1   |
| C      | 80  | 20  | 0   | 0   | 1   |
| D      | 80  | 20  | 0   | 0   | 1   |
| E      | 80  | 20  | 0   | 0   | 1   |
| F      | 100 | 0   | 0   | 0   | 1   |
| G      | 80  | 20  | 0   | 0   | 1   |
| H      | 80  | 20  | 0   | 0   | 1   |
| I      | 80  | 20  | 0   | 0   | 1   |
| J      | 20  | 60  | 0   | 20  | 1   |
| AB     | 20  | 60  | 0   | 20  | 2   |
| AC     | 0   | 100 | 0   | 0   | 2   |
| AD     | 0   | 39  | 0   | 61  | 2   |
| K      | 0   | 75  | 0   | 25  | 2   |
| L      | 0   | 39  | 0   | 61  | 2   |
| N      | 0   | 39  | 0   | 61  | 2   |
| P      | 0   | 39  | 0   | 61  | 2   |
| Q      | 0   | 39  | 0   | 61  | 2   |
| S      | 0   | 5   | 70  | 25  | 2   |
| U      | 80  | 20  | 0   | 0   | 2   |
| V      | 80  | 20  | 0   | 0   | 2   |
| X      | 0   | 75  | 0   | 25  | 2   |
| Y      | 20  | 60  | 0   | 20  | 2   |
| Z      | 20  | 60  | 0   | 20  | 2   |
| AF     | 0   | 5   | 70  | 25  | 3   |
| AH     | 0   | 5   | 70  | 25  | 3   |
| AL     | 0   | 5   | 70  | 25  | 3   |
| AM     | 0   | 5   | 70  | 25  | 3   |
| AO     | 0   | 0   | 100 | 0   | 3   |
| AP     | 0   | 5   | 70  | 25  | 3   |
| AR     | 0   | 16  | 67  | 17  | 3   |
| AU     | 0   | 16  | 67  | 17  | 3   |
| AW     | 0   | 5   | 70  | 25  | 3   |
| AX     | 0   | 5   | 70  | 25  | 3   |
| AZ     | 0   | 39  | 0   | 61  | 4   |
| BA_rot | 0   | 39  | 0   | 61  | 4   |
| BB     | 0   | 39  | 0   | 61  | 4   |
| BC     | 0   | 39  | 0   | 61  | 4   |
| BD     | 0   | 39  | 0   | 61  | 4   |
| BE     | 0   | 39  | 0   | 61  | 4   |
| BF     | 0   | 39  | 0   | 61  | 4   |
| BG     | 0   | 5   | 70  | 25  | 4   |
| BH     | 0   | 39  | 0   | 61  | 4   |
| BI     | 0   | 39  | 0   | 61  | 4   |
| BJ     | 20  | 60  | 0   | 20  | 4   |
| BK     | 0   | 0   | 0   | 100 | 4   |
| NM1    | 0   | 5   | 70  | 25  | 4   |
| NM2    | 0   | 16  | 67  | 17  | 4   |



fr\_aldehyde Removed

Test Set

Confusion Matrix

|       |           |            |            |            |             |
|-------|-----------|------------|------------|------------|-------------|
| 1     | 0<br>(0%) | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%)   |
| 2     | 0<br>(0%) | 2<br>(50%) | 0<br>(0%)  | 2<br>(50%) | 4<br>33.3%  |
| 3     | 0<br>(0%) | 1<br>12.5% | 7<br>87.5% | 0<br>(0%)  | 8<br>66.7%  |
| 4     | 0<br>(0%) | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%)   |
| total | 0<br>(0%) | 3<br>66.7% | 7<br>100%  | 2<br>(0%)  | 12<br>(75%) |
|       | 1         | 2          | 3          | 4          | total       |

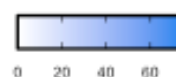
■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■

fr\_aldehyde Removed

## Test Set

### Probability Heatmap

|    | 1  | 2  | 3  | 4  | Exp |
|----|----|----|----|----|-----|
| M  | 0  | 39 | 0  | 61 | 2   |
| O  | 0  | 39 | 0  | 61 | 2   |
| W  | 0  | 75 | 0  | 25 | 2   |
| AK | 0  | 5  | 70 | 25 | 3   |
| AN | 0  | 5  | 70 | 25 | 3   |
| AV | 0  | 16 | 67 | 17 | 3   |
| T  | 0  | 75 | 0  | 25 | 2   |
| AT | 0  | 16 | 67 | 17 | 3   |
| AS | 0  | 16 | 67 | 17 | 3   |
| AQ | 20 | 60 | 0  | 20 | 3   |
| AI | 0  | 5  | 70 | 25 | 3   |
| AG | 0  | 5  | 70 | 25 | 3   |



fr\_aldehyde Removed

### fr\_amide Removed

formula

---

class ~ fr\_NH0 + fr\_aldehyde + fr\_pyridine

Full Model Stats - Overall Accuracy and Pseudo-R2

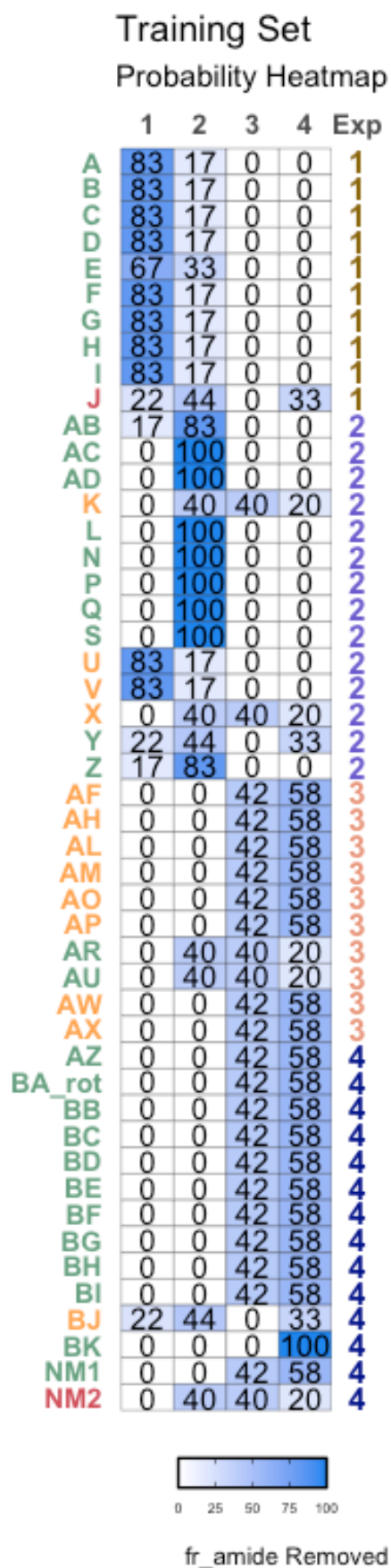
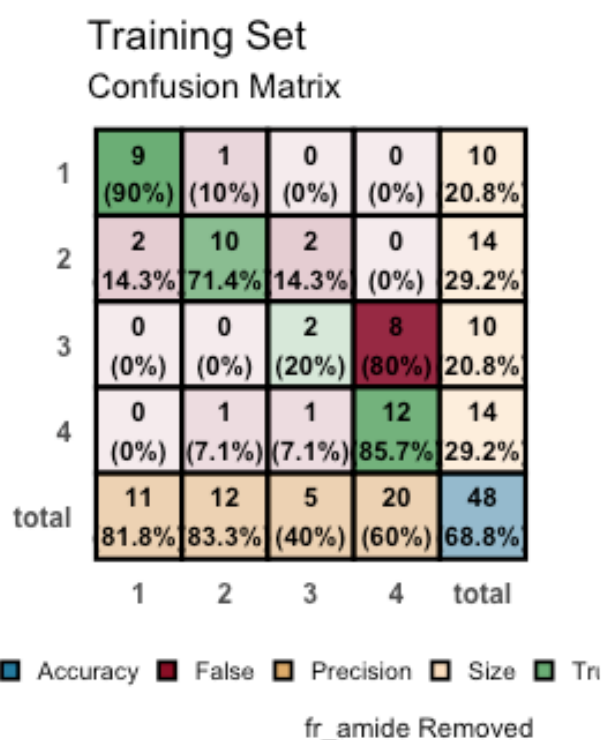
Accuracy    McFadden\_R2

---

68.75%    0.584

### Model Coefficients

|   | (Intercept) | fr_NH01 | fr_NH02 | fr_NH05 | fr_aldehyde1 | fr_pyridine1 | fr_pyridine2 |
|---|-------------|---------|---------|---------|--------------|--------------|--------------|
| 2 | 22.909      | 16.194  | -22.216 | -3.564  | 0.916        | -40.712      | 29.776       |
| 3 | 40.565      | -1.462  | -63.228 | -15.864 | -50.300      | -61.605      | -6.107       |
| 4 | 40.884      | -2.474  | -40.478 | 23.878  | -51.100      | -58.059      | -15.830      |



Test Set

Confusion Matrix

|       |           |            |            |            |             |
|-------|-----------|------------|------------|------------|-------------|
| 1     | 0<br>(0%) | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%)   |
| 2     | 0<br>(0%) | 2<br>(50%) | 2<br>(50%) | 0<br>(0%)  | 4<br>33.3%  |
| 3     | 0<br>(0%) | 1<br>12.5% | 3<br>37.5% | 4<br>(50%) | 8<br>66.7%  |
| 4     | 0<br>(0%) | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%)   |
| total | 0<br>(0%) | 3<br>66.7% | 5<br>(60%) | 4<br>(0%)  | 12<br>41.7% |
|       | 1         | 2          | 3          | 4          | total       |

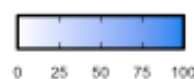
■ Accuracy
 ■ False
 ■ Precision
 ■ Size
 ■

fr\_amide Removed

## Test Set

### Probability Heatmap

|    | 1  | 2   | 3  | 4  | Exp |
|----|----|-----|----|----|-----|
| M  | 0  | 100 | 0  | 0  | 2   |
| O  | 0  | 100 | 0  | 0  | 2   |
| W  | 0  | 40  | 40 | 20 | 2   |
| AK | 0  | 0   | 42 | 58 | 3   |
| AN | 0  | 0   | 42 | 58 | 3   |
| AV | 0  | 40  | 40 | 20 | 3   |
| T  | 0  | 40  | 40 | 20 | 2   |
| AT | 0  | 40  | 40 | 20 | 3   |
| AS | 0  | 40  | 40 | 20 | 3   |
| AQ | 22 | 44  | 0  | 33 | 3   |
| AI | 0  | 0   | 42 | 58 | 3   |
| AG | 0  | 0   | 42 | 58 | 3   |



fr\_amide Removed

### fr\_pyridine Removed

formula

---

class ~ fr\_NH0 + fr\_aldehyde + fr\_amide

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy    McFadden\_R2

---

79.17%    0.676

Model Coefficients

|   | (Intercept) | fr_NH01 | fr_NH02 | fr_NH05 | fr_aldehyde1 | fr_amide1 | fr_amide2 |
|---|-------------|---------|---------|---------|--------------|-----------|-----------|
| 2 | 40.004      | -40.711 | -38.683 | -6.192  | 0.167        | -22.716   | -9.730    |
| 3 | 9.550       | -85.333 | -25.625 | -2.528  | -95.463      | 76.475    | 60.011    |
| 4 | 58.940      | -84.773 | -58.481 | 16.210  | -104.975     | 25.833    | -36.687   |

Training Set  
Confusion Matrix

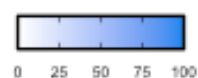
|       |             |             |             |             |             |
|-------|-------------|-------------|-------------|-------------|-------------|
| 1     | 8<br>(80%)  | 1<br>(10%)  | 1<br>(10%)  | 0<br>(0%)   | 10<br>20.8% |
| 2     | 4<br>28.6%  | 10<br>71.4% | 0<br>(0%)   | 0<br>(0%)   | 14<br>29.2% |
| 3     | 0<br>(0%)   | 0<br>(0%)   | 10<br>100%  | 0<br>(0%)   | 10<br>20.8% |
| 4     | 0<br>(0%)   | 1<br>(7.1%) | 3<br>21.4%  | 10<br>71.4% | 14<br>29.2% |
| total | 12<br>66.7% | 12<br>83.3% | 14<br>71.4% | 10<br>100%  | 48<br>79.2% |
|       | 1           | 2           | 3           | 4           | total       |

■ Accuracy ■ False ■ Precision ■ Size ■ Tr

fr\_pyridine Removed

Training Set  
Probability Heatmap

|        |    |     |     |     |     |
|--------|----|-----|-----|-----|-----|
|        | 1  | 2   | 3   | 4   | Exp |
| A      | 67 | 33  | 0   | 0   | 1   |
| B      | 67 | 33  | 0   | 0   | 1   |
| C      | 67 | 33  | 0   | 0   | 1   |
| D      | 67 | 33  | 0   | 0   | 1   |
| E      | 63 | 37  | 0   | 0   | 1   |
| F      | 25 | 0   | 50  | 25  | 1   |
| G      | 67 | 33  | 0   | 0   | 1   |
| H      | 67 | 33  | 0   | 0   | 1   |
| I      | 67 | 33  | 0   | 0   | 1   |
| J      | 16 | 59  | 0   | 25  | 1   |
| AB     | 18 | 82  | 0   | 0   | 2   |
| AC     | 16 | 59  | 0   | 25  | 2   |
| AD     | 0  | 100 | 0   | 0   | 2   |
| K      | 67 | 33  | 0   | 0   | 2   |
| L      | 0  | 100 | 0   | 0   | 2   |
| N      | 0  | 100 | 0   | 0   | 2   |
| P      | 0  | 100 | 0   | 0   | 2   |
| Q      | 0  | 100 | 0   | 0   | 2   |
| S      | 0  | 100 | 0   | 0   | 2   |
| U      | 67 | 33  | 0   | 0   | 2   |
| V      | 67 | 33  | 0   | 0   | 2   |
| X      | 67 | 33  | 0   | 0   | 2   |
| Y      | 16 | 59  | 0   | 25  | 2   |
| Z      | 18 | 82  | 0   | 0   | 2   |
| AF     | 0  | 0   | 78  | 22  | 3   |
| AH     | 0  | 0   | 78  | 22  | 3   |
| AL     | 0  | 0   | 78  | 22  | 3   |
| AM     | 0  | 0   | 78  | 22  | 3   |
| AO     | 0  | 0   | 100 | 0   | 3   |
| AP     | 0  | 0   | 78  | 22  | 3   |
| AR     | 25 | 0   | 50  | 25  | 3   |
| AU     | 25 | 0   | 50  | 25  | 3   |
| AW     | 0  | 0   | 78  | 22  | 3   |
| AX     | 0  | 0   | 78  | 22  | 3   |
| AZ     | 0  | 0   | 0   | 100 | 4   |
| BA_rot | 0  | 0   | 0   | 100 | 4   |
| BB     | 0  | 0   | 0   | 100 | 4   |
| BC     | 0  | 0   | 0   | 100 | 4   |
| BD     | 0  | 0   | 0   | 100 | 4   |
| BE     | 0  | 0   | 0   | 100 | 4   |
| BF     | 0  | 0   | 0   | 100 | 4   |
| BG     | 0  | 0   | 78  | 22  | 4   |
| BH     | 0  | 0   | 0   | 100 | 4   |
| BI     | 0  | 0   | 0   | 100 | 4   |
| BJ     | 16 | 59  | 0   | 25  | 4   |
| BK     | 0  | 0   | 0   | 100 | 4   |
| NM1    | 0  | 0   | 78  | 22  | 4   |
| NM2    | 25 | 0   | 50  | 25  | 4   |



fr\_pyridine Removed



Test Set

Confusion Matrix

|       |            |            |            |           |             |
|-------|------------|------------|------------|-----------|-------------|
| 1     | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%) | 0<br>(0%)   |
| 2     | 2<br>(50%) | 2<br>(50%) | 0<br>(0%)  | 0<br>(0%) | 4<br>33.3%  |
| 3     | 0<br>(0%)  | 1<br>12.5% | 7<br>87.5% | 0<br>(0%) | 8<br>66.7%  |
| 4     | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%)  | 0<br>(0%) | 0<br>(0%)   |
| total | 2<br>(0%)  | 3<br>66.7% | 7<br>100%  | 0<br>(0%) | 12<br>(75%) |
|       | 1          | 2          | 3          | 4         | total       |

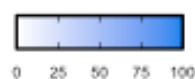
■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■

fr\_pyridine Removed

## Test Set

### Probability Heatmap

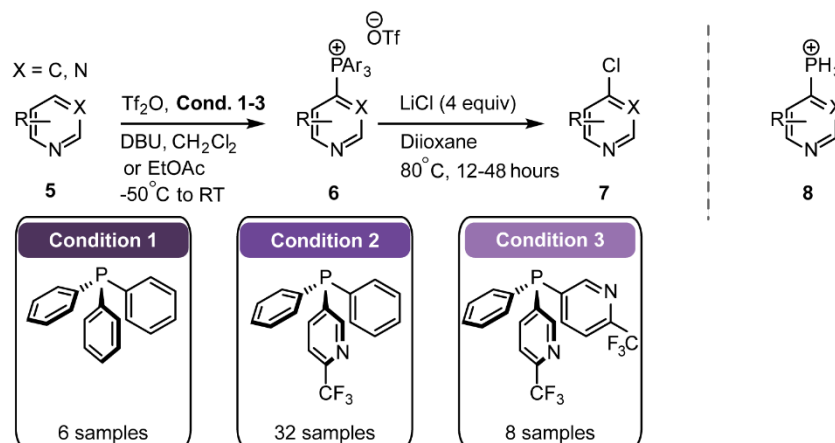
|    | 1  | 2   | 3  | 4  | Exp |
|----|----|-----|----|----|-----|
| M  | 0  | 100 | 0  | 0  | 2   |
| O  | 0  | 100 | 0  | 0  | 2   |
| W  | 67 | 33  | 0  | 0  | 2   |
| AK | 0  | 0   | 78 | 22 | 3   |
| AN | 0  | 0   | 78 | 22 | 3   |
| AV | 25 | 0   | 50 | 25 | 3   |
| T  | 67 | 33  | 0  | 0  | 2   |
| AT | 25 | 0   | 50 | 25 | 3   |
| AS | 25 | 0   | 50 | 25 | 3   |
| AQ | 16 | 59  | 0  | 25 | 3   |
| AI | 0  | 0   | 78 | 22 | 3   |
| AG | 0  | 0   | 78 | 22 | 3   |



fr\_pyridine Removed

## 2.3 Selective pyridine halogenation through phosphine functionalization

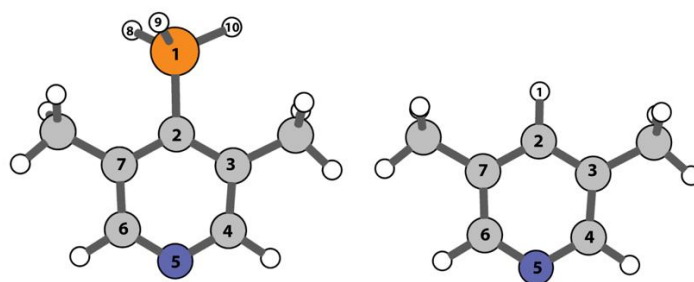
The following sections contain the list of best models as extracted from the code in R provided in our GitHub account. The scopes and reaction details are given in the main SI. The reaction to which the following model attributes is as follows:



**Figure S5:** Representation of the reaction

### 2.3.1 Features extracted for model identification

Information about the different RDKit fragments, their names and definitions are available on the modules [webpage](#). We screened out features that had no variance (all the same value), as well as those that their sum indicated that they represent too little examples. If, for instance, the sum of the feature is 6, and there are 60 data points, the feature describes 10% or less of the dataset. This was done in order to avoid a data set inflated by features with little variance.



| Feature             | Atom indices-<br>Pyridine (SM) | Atom indices-<br>Phosphinium<br>(P) | Dataset naming<br>scheme | Manuscript<br>naming scheme |
|---------------------|--------------------------------|-------------------------------------|--------------------------|-----------------------------|
| Sterimol<br>(along) | 5-2 & 7-4                      |                                     | B1_1 and B1_2            | $B_{1(5-2)}$ or $L_{7-4}$   |

|                                                |                                                                                        |                                                                     |                                                           |                                                          |
|------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|
| <b>NPA charges</b>                             | 1 through 7 and sum and average of 2 to 7                                              | 1, 2, 4, 5 and average of H atoms on P                              | NPA_1<br>or<br>NPA_Avg_H_on P<br>NPA_sum_SM<br>Avg_NPA_SM | $q_1$<br>$\bar{q}_{Ar}$                                  |
| <b>NPA charge differences</b>                  | Difference between C and H going through halogenation                                  |                                                                     | diff_NBO_C_Hhal                                           | $\Delta q_{C-H}$                                         |
| <b>Dipole moment and its components</b>        | Origin of coordinates at 5, y direction 2 and xy plane on the ring                     | Origin of coordinates at 5, y direction 2 and xy plane on the ring  | Total<br>dip_x<br>dip_y<br>dip_z                          | $\mu_{tot}$<br>$\mu_x$<br>$\mu_y$<br>$\mu_z$             |
| <b>Bond stretch frequency</b>                  | None                                                                                   | Max, min and avg frequency of P-H                                   | max_PH_vib<br>min_PH_vib<br>Avg_PH_vib                    | $\nu_{P-H(max)}$<br>$\nu_{P-H(min)}$<br>$\nu_{P-H(Avg)}$ |
| <b>Ring vibrations</b>                         | Pyridine ring                                                                          | Pyridine ring                                                       | Para & cross                                              | $\nu_{cross}$<br>$\nu_{para}$                            |
| <b>Bond lengths and distance between atoms</b> | Distance H going through halogenation and carbon connected to it, average distance P-H | Distance phosphine and carbon connected to it, average distance P-H | Dist_(P, C)_p<br>Avg_dist_PH                              | $d_{1-2}$<br>$d_{P-C}$                                   |
| <b>Polarisability</b>                          | Isotropic and Anisotropic polarizability                                               | Anisotropic polarizability                                          | Iso.Polar_SM<br>Aniso.Polar_P                             | Iso. Pol<br>Aniso. Pol                                   |

**Figure S6 and Table S3:** Representation of the atom indexing and Gaussian calculation's derived feature set

### **2.3.2 Best model and model identification reports – binary viability model**

Below is the data for the model presented in the main text as it appears in the output generated by our statistical code.

Below are the confusion matrices and probability heatmaps for the training and test sets, as they result from our code.

The most predictive binary viability model showed an accuracy of 93% for the training set and of 88% for the test set (representing only three errors out of 33 examples). The features comprising this model were the fragments of methyl aryls, nitrogen-containing aryls, benzenes, and ethers. Based on this and the previous case study, we strongly recommend testing fragments as features in condition classification algorithms because of their negligible computational cost.

In the following section – class 1 refers to the reactive molecules, and class 2 refers to unreactive molecules or the “null” category

| formula                                                                       | McFadden<br>R2 |
|-------------------------------------------------------------------------------|----------------|
| `class` ~ `fr_Ar_N` + `fr_aryl_methyl` + `fr_benzene` + `fr_halogen`          | 0.721          |
| `class` ~ `fr_aryl_methyl` + `fr_benzene` + `fr_halogen` +<br>`fr_pyridine`   | 0.721          |
| `class` ~ `NPA_1_P` + `fr_Ar_N` + `fr_aryl_methyl` + `fr_benzene`             | 0.689          |
| `class` ~ `NPA_1_P` + `B1_2_SM` + `Iso.Polar_SM` + `fr_Ar_N`                  | 0.683          |
| `class` ~ `cross_P` + `fr_Ar_N` + `fr_aryl_methyl` + `fr_halogen`             | 0.676          |
| `class` ~ `cross_P` + `fr_aryl_methyl` + `fr_halogen` + `fr_pyridine`         | 0.676          |
| `class` ~ `NPA_1_P` + `Iso.Polar_SM` + `fr_Ar_N` + `fr_ether`                 | 0.667          |
| <b>`class` ~ `fr_Ar_N` + `fr_aryl_methyl` + `fr_benzene` + `fr_ether`</b>     | <b>0.662</b>   |
| `class` ~ `NPA_5_P` + `fr_Ar_N` + `fr_aryl_methyl` + `fr_benzene`             | 0.657          |
| `class` ~ `NPA_Avg_H_onPhos` + `fr_Ar_N` + `fr_aryl_methyl` +<br>`fr_benzene` | 0.644          |

### **Published Model**

Full Model Stats - Overall Accuracy and Pseudo-R2

| Accuracy | McFadden_R2 |
|----------|-------------|
| 93.02%   | 0.662       |

Model Coefficients

|                 |          |
|-----------------|----------|
| (Intercept)     | 26.65774 |
| fr_Ar_N1        | -24.7479 |
| fr_Ar_N2        | -61.5889 |
| fr_Ar_N3        | -40.4121 |
| fr_aryl_methyl1 | -3.58992 |
| fr_aryl_methyl2 | -24.2613 |
| fr_benzene1     | -3.72567 |

|             |          |
|-------------|----------|
| fr_benzene2 | -39.3729 |
| fr_ether1   | 3.239299 |
| fr_ether2   | -12.4055 |

| Training Set |             |             |             |
|--------------|-------------|-------------|-------------|
| 1            | 30<br>(97%) | 1<br>(3%)   | 31<br>(72%) |
| 2            | 2<br>(17%)  | 10<br>(83%) | 12<br>(28%) |
| total        | 32<br>(94%) | 11<br>(91%) | 43<br>(93%) |
|              | 1           | 2           | total       |

■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

8. 8th Place

Training Set  
Probability Heatmap

|    | 1   | 2   | Exp |
|----|-----|-----|-----|
| A  | 100 | 0   | 1   |
| AF | 84  | 16  | 1   |
| B  | 86  | 14  | 1   |
| D  | 100 | 0   | 1   |
| E  | 100 | 0   | 1   |
| F  | 100 | 0   | 1   |
| AC | 86  | 14  | 1   |
| AD | 100 | 0   | 1   |
| AE | 100 | 0   | 1   |
| AH | 84  | 16  | 1   |
| AI | 100 | 0   | 1   |
| AL | 100 | 0   | 1   |
| AM | 84  | 16  | 1   |
| H  | 13  | 87  | 1   |
| K  | 84  | 16  | 1   |
| N  | 100 | 0   | 1   |
| O  | 86  | 14  | 1   |
| P  | 100 | 0   | 1   |
| R  | 100 | 0   | 1   |
| S  | 100 | 0   | 1   |
| T  | 100 | 0   | 1   |
| U  | 90  | 10  | 1   |
| V  | 86  | 14  | 1   |
| W  | 100 | 0   | 1   |
| X  | 100 | 0   | 1   |
| Y  | 100 | 0   | 1   |
| Z  | 90  | 10  | 1   |
| AN | 86  | 14  | 1   |
| AO | 100 | 0   | 1   |
| AP | 86  | 14  | 1   |
| AS | 100 | 0   | 1   |
| AV | 1   | 99  | 2   |
| AW | 13  | 87  | 2   |
| AX | 84  | 16  | 2   |
| AY | 13  | 87  | 2   |
| AZ | 1   | 99  | 2   |
| BA | 13  | 87  | 2   |
| BB | 13  | 87  | 2   |
| BC | 13  | 87  | 2   |
| BE | 0   | 100 | 2   |
| BF | 0   | 100 | 2   |
| BH | 86  | 14  | 2   |
| BI | 19  | 81  | 2   |



8. 8th Place

### Test Set

|       |            |            |             |       |
|-------|------------|------------|-------------|-------|
|       |            | 1          | 2           | total |
| 1     | 8<br>(80%) | 2<br>(20%) | 10<br>(71%) |       |
| 2     | 1<br>(25%) | 3<br>(75%) | 4<br>(29%)  |       |
| total | 9<br>(89%) | 5<br>(60%) | 14<br>(79%) |       |
|       | 1          | 2          | total       |       |

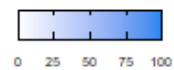
■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

8. 8th Place

### Test Set

Probability Heatmap

|    | 1   | 2  | Exp |
|----|-----|----|-----|
| AA | 100 | 0  | 1   |
| AB | 1   | 99 | 1   |
| AJ | 100 | 0  | 1   |
| AK | 84  | 16 | 1   |
| G  | 1   | 99 | 1   |
| I  | 86  | 14 | 1   |
| M  | 86  | 14 | 1   |
| AQ | 100 | 0  | 1   |
| AT | 100 | 0  | 1   |
| AU | 86  | 14 | 1   |
| BD | 1   | 99 | 2   |
| BG | 86  | 14 | 2   |
| BJ | 1   | 99 | 2   |
| BK | 19  | 81 | 2   |



8. 8th Place



### **Model Analysis - Variable Impact**

To evaluate the importance of each of the features, we analyzed the loss in accuracy by removing each of them, similarly to the previous case studies. Omitting the nitrogen containing aryl or benzene fragments significantly decreased the ability of the model to classify substrates to the “null” class correctly. Omitting the methyl aryls decreased the ability of the model to classify substrates to the “viable” category. Omitting the ether fragment has the least impact on the model’s accuracy but still improves its ability to distinguish the “null” class properly. Overall, it seems that taken together, these features lead to a strong viability classification, which highlights the unique value of using an algorithm to identify hidden, unintuitive patterns in the data.

To understand the analysis made in the SI, it is recommended to note the recall (marked in green in the confusion table) and the accuracy (marked in blue in each confusion table), and compare it back to the original confusion table of the chosen model shown above. This comparison will illustrate the impact each feature has on the model.

fr\_Ar\_N Removed

formula

---

class ~ fr\_aryl\_methyl + fr\_benzene + fr\_ether

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy    McFadden\_R2

---

69.77%    0.186

Model Coefficients

|                 | x        |
|-----------------|----------|
| (Intercept)     | -0.01961 |
| fr_aryl_methyl1 | -1.90334 |
| fr_aryl_methyl2 | -9.389   |
| fr_benzene1     | -1.28214 |
| fr_benzene2     | -9.37829 |
| fr_ether1       | 0.186187 |
| fr_ether2       | 0.519278 |

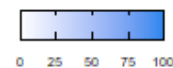
| Training Set |             |            |             |
|--------------|-------------|------------|-------------|
| 1            | 28<br>(90%) | 3<br>(10%) | 31<br>(72%) |
| 2            | 10<br>(83%) | 2<br>(17%) | 12<br>(28%) |
| total        | 38<br>(74%) | 5<br>(40%) | 43<br>(70%) |
|              | 1           | 2          | total       |

■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

8. 8th Place

Training Set  
Probability Heatmap

|    | 1   | 2  | Exp |
|----|-----|----|-----|
| A  | 100 | 0  | 1   |
| AF | 87  | 13 | 1   |
| B  | 79  | 21 | 1   |
| D  | 75  | 25 | 1   |
| E  | 100 | 0  | 1   |
| F  | 46  | 54 | 1   |
| AC | 79  | 21 | 1   |
| AD | 100 | 0  | 1   |
| AE | 75  | 25 | 1   |
| AH | 87  | 13 | 1   |
| AI | 46  | 54 | 1   |
| AL | 96  | 4  | 1   |
| AM | 87  | 13 | 1   |
| H  | 50  | 50 | 1   |
| K  | 87  | 13 | 1   |
| N  | 50  | 50 | 1   |
| O  | 79  | 21 | 1   |
| P  | 50  | 50 | 1   |
| R  | 87  | 13 | 1   |
| S  | 50  | 50 | 1   |
| T  | 50  | 50 | 1   |
| U  | 95  | 5  | 1   |
| V  | 79  | 21 | 1   |
| W  | 94  | 6  | 1   |
| X  | 79  | 21 | 1   |
| Y  | 96  | 4  | 1   |
| Z  | 95  | 5  | 1   |
| AN | 79  | 21 | 1   |
| AO | 38  | 62 | 1   |
| AP | 79  | 21 | 1   |
| AS | 100 | 0  | 1   |
| AV | 46  | 54 | 2   |
| AW | 50  | 50 | 2   |
| AX | 87  | 13 | 2   |
| AY | 50  | 50 | 2   |
| AZ | 46  | 54 | 2   |
| BA | 50  | 50 | 2   |
| BB | 50  | 50 | 2   |
| BC | 50  | 50 | 2   |
| BE | 50  | 50 | 2   |
| BF | 69  | 31 | 2   |
| BH | 79  | 21 | 2   |
| BI | 75  | 25 | 2   |



8. 8th Place

**Test Set**

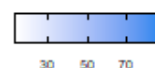
|       |            |            |             |
|-------|------------|------------|-------------|
|       |            |            |             |
| 1     | 6<br>(60%) | 4<br>(40%) | 10<br>(71%) |
| 2     | 2<br>(50%) | 2<br>(50%) | 4<br>(29%)  |
| total | 8<br>(75%) | 6<br>(33%) | 14<br>(57%) |
|       | 1          | 2          | total       |

■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

8. 8th Place

**Test Set**  
Probability Heatmap

|    |    |    |     |
|----|----|----|-----|
|    | 1  | 2  | Exp |
| AA | 38 | 62 | 1   |
| AB | 46 | 54 | 1   |
| AJ | 50 | 50 | 1   |
| AK | 87 | 13 | 1   |
| G  | 46 | 54 | 1   |
| I  | 79 | 21 | 1   |
| M  | 79 | 21 | 1   |
| AQ | 69 | 31 | 1   |
| AT | 46 | 54 | 1   |
| AU | 79 | 21 | 1   |
| BD | 46 | 54 | 2   |
| BG | 79 | 21 | 2   |
| BJ | 46 | 54 | 2   |
| BK | 75 | 25 | 2   |



8. 8th Place

### fr\_aryl\_methyl Removed

formula

---

class ~ fr\_Ar\_N + fr\_benzene + fr\_ether

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy    McFadden\_R2

---

81.4%        0.463

Model Coefficients

|             | x        |
|-------------|----------|
| (Intercept) | 18.48661 |
| fr_Ar_N1    | -18.4119 |
| fr_Ar_N2    | -29.8097 |
| fr_Ar_N3    | -29.6786 |
| fr_benzene1 | -2.43604 |
| fr_benzene2 | -28.9911 |
| fr_ether1   | 1.990374 |
| fr_ether2   | -7.98677 |

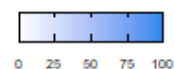
| Training Set |             |             |             |
|--------------|-------------|-------------|-------------|
| 1            | 25<br>(81%) | 6<br>(19%)  | 31<br>(72%) |
| 2            | 2<br>(17%)  | 10<br>(83%) | 12<br>(28%) |
| total        | 27<br>(93%) | 16<br>(62%) | 43<br>(81%) |
|              | 1           | 2           | total       |

■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

8. 8th Place

Training Set  
Probability Heatmap

|    | 1   | 2   | Exp |
|----|-----|-----|-----|
| A  | 48  | 52  | 1   |
| AF | 48  | 52  | 1   |
| B  | 91  | 9   | 1   |
| D  | 100 | 0   | 1   |
| E  | 100 | 0   | 1   |
| F  | 100 | 0   | 1   |
| AC | 91  | 9   | 1   |
| AD | 100 | 0   | 1   |
| AE | 100 | 0   | 1   |
| AH | 48  | 52  | 1   |
| AI | 100 | 0   | 1   |
| AL | 91  | 9   | 1   |
| AM | 48  | 52  | 1   |
| H  | 48  | 52  | 1   |
| K  | 48  | 52  | 1   |
| N  | 100 | 0   | 1   |
| O  | 91  | 9   | 1   |
| P  | 100 | 0   | 1   |
| R  | 100 | 0   | 1   |
| S  | 100 | 0   | 1   |
| T  | 100 | 0   | 1   |
| U  | 59  | 41  | 1   |
| V  | 91  | 9   | 1   |
| W  | 100 | 0   | 1   |
| X  | 100 | 0   | 1   |
| Y  | 91  | 9   | 1   |
| Z  | 59  | 41  | 1   |
| AN | 91  | 9   | 1   |
| AO | 100 | 0   | 1   |
| AP | 91  | 9   | 1   |
| AS | 100 | 0   | 1   |
| AV | 11  | 89  | 2   |
| AW | 48  | 52  | 2   |
| AX | 48  | 52  | 2   |
| AY | 48  | 52  | 2   |
| AZ | 11  | 89  | 2   |
| BA | 48  | 52  | 2   |
| BB | 48  | 52  | 2   |
| BC | 48  | 52  | 2   |
| BE | 0   | 100 | 2   |
| BF | 0   | 100 | 2   |
| BH | 91  | 9   | 2   |
| BI | 59  | 41  | 2   |



8. 8th Place

**Test Set**

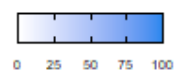
|       |            |            |             |
|-------|------------|------------|-------------|
|       | 1          | 2          |             |
| 1     | 7<br>(70%) | 3<br>(30%) | 10<br>(71%) |
| 2     | 2<br>(50%) | 2<br>(50%) | 4<br>(29%)  |
| total | 9<br>(78%) | 5<br>(40%) | 14<br>(64%) |
|       | 1          | 2          | total       |

■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

8. 8th Place

**Test Set**  
Probability Heatmap

|    |     |    |     |
|----|-----|----|-----|
|    | 1   | 2  | Exp |
| AA | 100 | 0  | 1   |
| AB | 11  | 89 | 1   |
| AJ | 100 | 0  | 1   |
| AK | 48  | 52 | 1   |
| G  | 11  | 89 | 1   |
| I  | 91  | 9  | 1   |
| M  | 91  | 9  | 1   |
| AQ | 100 | 0  | 1   |
| AT | 100 | 0  | 1   |
| AU | 91  | 9  | 1   |
| BD | 11  | 89 | 2   |
| BG | 91  | 9  | 2   |
| BJ | 11  | 89 | 2   |
| BK | 59  | 41 | 2   |



8. 8th Place

### fr\_benzene Removed

formula

---

class ~ Avg\_NPA\_SM + fr\_Ar\_N + fr\_aryl\_methyl + fr\_halogen

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy    McFadden\_R2

---

80.43%    0.404

Model Coefficients

|                 | x       |
|-----------------|---------|
| (Intercept)     | 15.336  |
| Avg_NPA_SM      | 19.324  |
| fr_Ar_N1        | -13.739 |
| fr_Ar_N2        | -26.002 |
| fr_Ar_N3        | -34.645 |
| fr_aryl_methyl1 | -2.552  |
| fr_aryl_methyl2 | -17.813 |
| fr_halogen1     | 1.004   |
| fr_halogen2     | 0.497   |
| fr_halogen3     | 0.231   |
| fr_halogen4     | -2.397  |
| fr_halogen5     | 0.961   |
| fr_halogen6     | -8.030  |



### Training Set

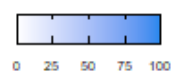
|       |             |            |             |
|-------|-------------|------------|-------------|
|       |             |            |             |
|       | 1           | 2          | total       |
| 1     | 29<br>(94%) | 2<br>(6%)  | 31<br>(72%) |
| 2     | 7<br>(58%)  | 5<br>(42%) | 12<br>(28%) |
| total | 36<br>(81%) | 7<br>(71%) | 43<br>(79%) |

■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

8. 8th Place

Training Set  
Probability Heatmap

|    | 1   | 2   | Exp |
|----|-----|-----|-----|
| A  | 100 | 0   | 1   |
| AF | 89  | 11  | 1   |
| B  | 52  | 48  | 1   |
| D  | 100 | 0   | 1   |
| E  | 100 | 0   | 1   |
| F  | 100 | 0   | 1   |
| AC | 52  | 48  | 1   |
| AD | 45  | 55  | 1   |
| AE | 100 | 0   | 1   |
| AH | 89  | 11  | 1   |
| AI | 100 | 0   | 1   |
| AL | 89  | 11  | 1   |
| AM | 89  | 11  | 1   |
| H  | 52  | 48  | 1   |
| K  | 89  | 11  | 1   |
| N  | 100 | 0   | 1   |
| O  | 52  | 48  | 1   |
| P  | 100 | 0   | 1   |
| R  | 100 | 0   | 1   |
| S  | 100 | 0   | 1   |
| T  | 100 | 0   | 1   |
| U  | 87  | 13  | 1   |
| V  | 52  | 48  | 1   |
| W  | 100 | 0   | 1   |
| X  | 100 | 0   | 1   |
| Y  | 89  | 11  | 1   |
| Z  | 87  | 13  | 1   |
| AN | 52  | 48  | 1   |
| AO | 100 | 0   | 1   |
| AP | 52  | 48  | 1   |
| AS | 45  | 55  | 1   |
| AV | 45  | 55  | 2   |
| AW | 52  | 48  | 2   |
| AX | 89  | 11  | 2   |
| AY | 52  | 48  | 2   |
| AZ | 45  | 55  | 2   |
| BA | 52  | 48  | 2   |
| BB | 52  | 48  | 2   |
| BC | 52  | 48  | 2   |
| BE | 0   | 100 | 2   |
| BF | 0   | 100 | 2   |
| BH | 52  | 48  | 2   |
| BI | 45  | 55  | 2   |



8. 8th Place

## Training Set

|       |             |            |             |
|-------|-------------|------------|-------------|
| 1     | 29<br>(94%) | 2<br>(6%)  | 31<br>(72%) |
| 2     | 7<br>(58%)  | 5<br>(42%) | 12<br>(28%) |
| total | 36<br>(81%) | 7<br>(71%) | 43<br>(79%) |
|       | 1           | 2          | total       |

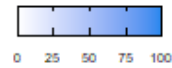
■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

8. 8th Place

## Test Set

Probability Heatmap

|    | 1   | 2  | Exp |
|----|-----|----|-----|
| AA | 100 | 0  | 1   |
| AB | 45  | 55 | 1   |
| AJ | 100 | 0  | 1   |
| AK | 89  | 11 | 1   |
| G  | 45  | 55 | 1   |
| I  | 52  | 48 | 1   |
| M  | 52  | 48 | 1   |
| AQ | 100 | 0  | 1   |
| AT | 100 | 0  | 1   |
| AU | 52  | 48 | 1   |
| BD | 45  | 55 | 2   |
| BG | 52  | 48 | 2   |
| BJ | 45  | 55 | 2   |
| BK | 45  | 55 | 2   |



8. 8th Place

### fr\_ether Removed

formula

---

class ~ Avg\_NPA\_SM + fr\_Ar\_N + fr\_aryl\_methyl + fr\_benzene

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy    McFadden\_R2

---

88.37%      0.534

Model Coefficients

|                 | x        |
|-----------------|----------|
| (Intercept)     | 11.23045 |
| fr_Ar_N1        | -9.9177  |
| fr_Ar_N2        | -22.5163 |
| fr_Ar_N3        | -22.4726 |
| fr_aryl_methyl1 | -2.81814 |
| fr_aryl_methyl2 | -15.9974 |
| fr_benzene1     | -2.47346 |
| fr_benzene2     | -21.0372 |

| Training Set |             |             |             |
|--------------|-------------|-------------|-------------|
| 1            | 29<br>(94%) | 2<br>(6%)   | 31<br>(72%) |
| 2            | 3<br>(25%)  | 9<br>(75%)  | 12<br>(28%) |
| total        | 32<br>(91%) | 11<br>(82%) | 43<br>(88%) |
|              | 1           | 2           | total       |

■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

8. 8th Place

Training Set  
Probability Heatmap

|    | 1   | 2   | Exp |
|----|-----|-----|-----|
| A  | 100 | 0   | 1   |
| AF | 82  | 18  | 1   |
| B  | 76  | 24  | 1   |
| D  | 100 | 0   | 1   |
| E  | 100 | 0   | 1   |
| F  | 100 | 0   | 1   |
| AC | 76  | 24  | 1   |
| AD | 100 | 0   | 1   |
| AE | 100 | 0   | 1   |
| AH | 82  | 18  | 1   |
| AI | 100 | 0   | 1   |
| AL | 98  | 2   | 1   |
| AM | 82  | 18  | 1   |
| H  | 21  | 79  | 1   |
| K  | 82  | 18  | 1   |
| N  | 100 | 0   | 1   |
| O  | 76  | 24  | 1   |
| P  | 100 | 0   | 1   |
| R  | 100 | 0   | 1   |
| S  | 100 | 0   | 1   |
| T  | 100 | 0   | 1   |
| U  | 98  | 2   | 1   |
| V  | 76  | 24  | 1   |
| W  | 98  | 2   | 1   |
| X  | 100 | 0   | 1   |
| Y  | 98  | 2   | 1   |
| Z  | 98  | 2   | 1   |
| AN | 76  | 24  | 1   |
| AO | 21  | 79  | 1   |
| AP | 76  | 24  | 1   |
| AS | 100 | 0   | 1   |
| AV | 21  | 79  | 2   |
| AW | 21  | 79  | 2   |
| AX | 82  | 18  | 2   |
| AY | 21  | 79  | 2   |
| AZ | 21  | 79  | 2   |
| BA | 21  | 79  | 2   |
| BB | 21  | 79  | 2   |
| BC | 21  | 79  | 2   |
| BE | 0   | 100 | 2   |
| BF | 0   | 100 | 2   |
| BH | 76  | 24  | 2   |
| BI | 76  | 24  | 2   |



8. 8th Place

**Test Set**

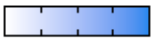
|       |            |            |             |
|-------|------------|------------|-------------|
|       | 1          | 2          |             |
| 1     | 7<br>(70%) | 3<br>(30%) | 10<br>(71%) |
| 2     | 2<br>(50%) | 2<br>(50%) | 4<br>(29%)  |
| total | 9<br>(78%) | 5<br>(40%) | 14<br>(64%) |
|       | 1          | 2          | total       |

■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

8. 8th Place

**Test Set**  
Probability Heatmap

|    | 1   | 2  | Exp |
|----|-----|----|-----|
| AA | 21  | 79 | 1   |
| AB | 21  | 79 | 1   |
| AJ | 100 | 0  | 1   |
| AK | 82  | 18 | 1   |
| G  | 21  | 79 | 1   |
| I  | 76  | 24 | 1   |
| M  | 76  | 24 | 1   |
| AQ | 76  | 24 | 1   |
| AT | 100 | 0  | 1   |
| AU | 76  | 24 | 1   |
| BD | 21  | 79 | 2   |
| BG | 76  | 24 | 2   |
| BJ | 21  | 79 | 2   |
| BK | 76  | 24 | 2   |



0 25 50 75 100

8. 8th Place

### 2.3.3 Best model and model identification reports – ordinal multinomial

#### logistic regression

Below is the data for the 3-category classification model of reactive substrates presented in the main text as it appears in the output generated by our statistical code.

| formula                                                    | McFadden R2  |
|------------------------------------------------------------|--------------|
| `class` ~ `para_P` + `NPA_6_SM` + `fr_aryl_methyl`         | 0.638        |
| `class` ~ `NPA_1_P` + `loc.B5_1_SM` + `fr_bicyclic`        | 0.616        |
| <b>`class` ~ `Dist_.P.C._P` + `NPA_6_SM` + `para_SM`</b>   | <b>0.607</b> |
| `class` ~ `NPA_6_SM` + `para_SM` + `fr_aryl_methyl`        | 0.607        |
| `class` ~ `NPA_5_P` + `Dist_.P.C._P` + `fr_aryl_methyl`    | 0.601        |
| `class` ~ `NPA_1_P` + `Total_P` + `fr_bicyclic`            | 0.597        |
| `class` ~ `Dist_.P.C._P` + `loc.B1_2_SM` + `para.angle_SM` | 0.573        |
| `class` ~ `para_P` + `NPA_6_SM` + `NPA_sum_SM`             | 0.569        |
| `class` ~ `NPA_6_SM` + `NPA_sum_SM` + `para_SM`            | 0.568        |
| `class` ~ `NPA_1_P` + `dip_y_P` + `fr_bicyclic`            | 0.563        |

#### **Published model**

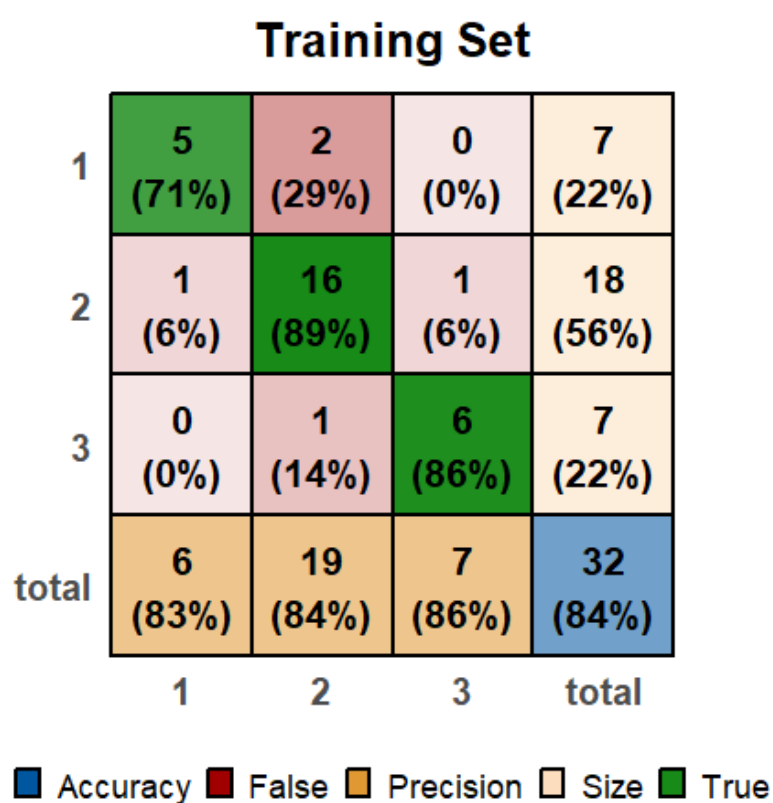
Full Model Stats - Overall Accuracy and Pseudo-R2

| Accuracy | McFadden_R2 |
|----------|-------------|
|----------|-------------|

|        |       |
|--------|-------|
| 84.38% | 0.607 |
|--------|-------|

Model Coefficients

|              | x            |
|--------------|--------------|
| Dist_(P,C)_P | -129.9237128 |
| NPA_6_SM     | 28.2302879   |
| para_SM      | 0.1585847    |

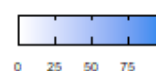


3. 3rd Place

### Training Set

Probability Heatmap

|    | 1  | 2  | 3  | Exp |
|----|----|----|----|-----|
| A  | 40 | 59 | 0  | 1   |
| AF | 78 | 22 | 0  | 1   |
| B  | 74 | 26 | 0  | 1   |
| C  | 50 | 49 | 0  | 1   |
| D  | 96 | 4  | 0  | 1   |
| E  | 41 | 59 | 0  | 1   |
| F  | 99 | 1  | 0  | 1   |
| AA | 20 | 80 | 0  | 2   |
| AB | 31 | 69 | 0  | 2   |
| AC | 18 | 82 | 0  | 2   |
| AE | 1  | 92 | 7  | 2   |
| AI | 2  | 95 | 3  | 2   |
| AK | 0  | 19 | 81 | 2   |
| AL | 2  | 95 | 3  | 2   |
| H  | 25 | 75 | 0  | 2   |
| I  | 15 | 85 | 0  | 2   |
| J  | 2  | 95 | 3  | 2   |
| L  | 1  | 90 | 9  | 2   |
| M  | 0  | 86 | 14 | 2   |
| S  | 58 | 42 | 0  | 2   |
| T  | 0  | 83 | 17 | 2   |
| V  | 40 | 60 | 0  | 2   |
| W  | 0  | 75 | 25 | 2   |
| Y  | 3  | 95 | 3  | 2   |
| Z  | 2  | 95 | 4  | 2   |
| AN | 0  | 13 | 87 | 3   |
| AP | 0  | 4  | 96 | 3   |
| AQ | 2  | 94 | 4  | 3   |
| AR | 0  | 9  | 91 | 3   |
| AS | 0  | 25 | 75 | 3   |
| AT | 0  | 16 | 84 | 3   |
| AU | 0  | 8  | 92 | 3   |



3. 3rd Place

## Test Set

|       |           |             |             |             |
|-------|-----------|-------------|-------------|-------------|
| 1     | 0<br>(0%) | 0<br>(0%)   | 0<br>(0%)   | 0<br>(0%)   |
| 2     | 0<br>(0%) | 8<br>(100%) | 0<br>(0%)   | 8<br>(89%)  |
| 3     | 0<br>(0%) | 0<br>(0%)   | 1<br>(100%) | 1<br>(11%)  |
| total | 0<br>(0%) | 8<br>(100%) | 1<br>(100%) | 9<br>(100%) |
|       | 1         | 2           | 3           | total       |

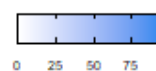
■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

3. 3rd Place

## Test Set

### Probability Heatmap

|    | 1  | 2  | 3  | Exp |
|----|----|----|----|-----|
| AH | 3  | 95 | 2  | 2   |
| AJ | 22 | 77 | 0  | 2   |
| G  | 39 | 61 | 0  | 2   |
| O  | 1  | 94 | 5  | 2   |
| P  | 6  | 93 | 1  | 2   |
| R  | 28 | 72 | 0  | 2   |
| U  | 18 | 81 | 0  | 2   |
| X  | 29 | 71 | 0  | 2   |
| AO | 0  | 10 | 89 | 3   |



3. 3rd Place



## Model Analysis - Variable Impact

To evaluate the importance of each of the features, we analyzed the loss in accuracy by removing each of them, similarly to the previous case studies. Omitting one of the characteristic 6-membered aromatic ring's vibrational mode on the starting material (para\_SM) had the largest impact on the model accuracy and is presumed to reflect both electronic and steric influences across the ring. Omitting the phosphorus–carbon bond length (Dist\_(P,C)\_P) and the NPA charge on ring atom #6, which is *ortho* to the nitrogen on the most substituted side of the molecule (vis-à-vis the phosphorus–nitrogen axis), both mildly impact the overall accuracy of the model. Dist\_(P,C)\_P particularly impacts the model ability to distinguish substrates that belong to Condition 1, whereas the NPA charge on ring atom #6 impacts the model's ability to properly classify substrates to Conditions 1 and 3.

To understand the analysis made in the SI, it is recommended to note the recall (marked in green in the confusion table) and the accuracy (marked in blue in each confusion table) and compare it back to the original confusion table of the chosen model shown above. This comparison will illustrate the impact each feature has on the model.

### Dist\_(P,C)\_P Removed

formula

---

class ~ NPA\_6\_SM + para\_SM

Full Model Stats - Overall Accuracy and Pseudo-R2

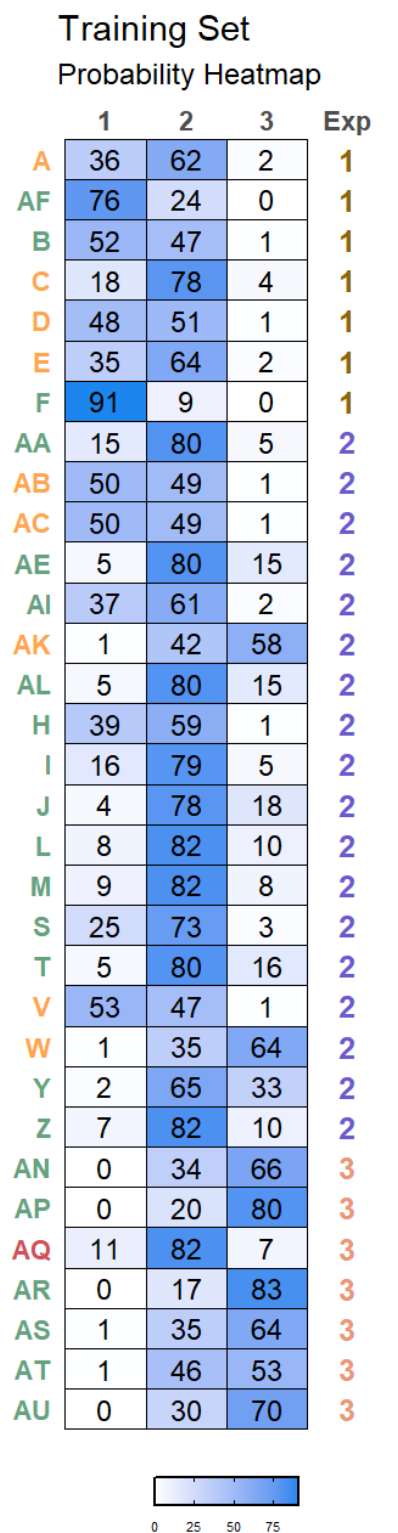
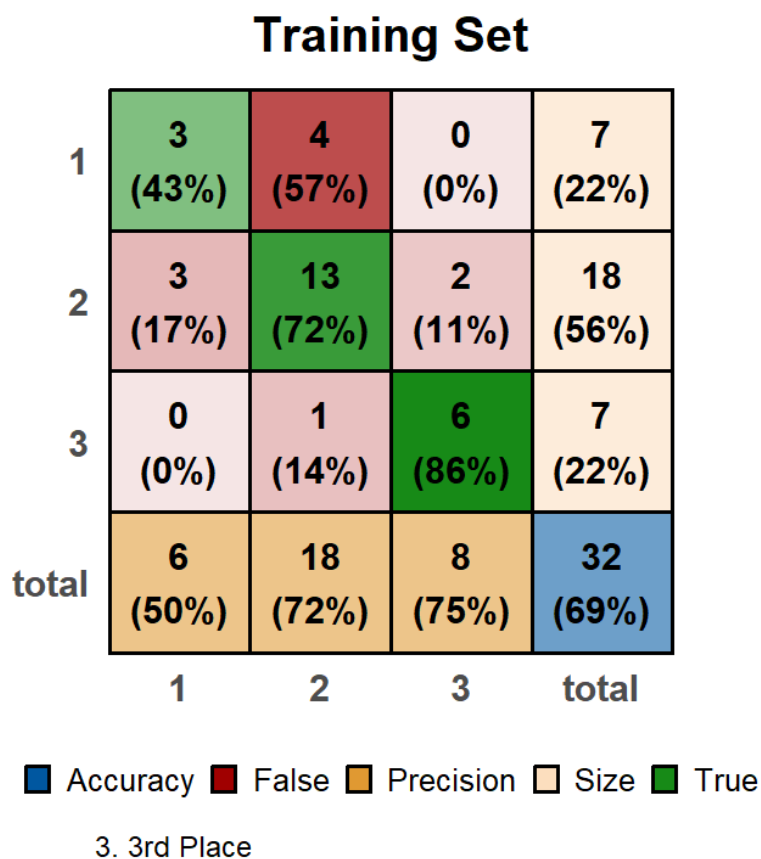
| Accuracy | McFadden_R2 |
|----------|-------------|
|----------|-------------|

---

|        |       |
|--------|-------|
| 68.75% | 0.422 |
|--------|-------|

Model Coefficients

|          | x          |
|----------|------------|
| NPA_6_SM | 22.5429404 |
| para_SM  | 0.0882469  |



## Test Set

|       |           |             |             |             |
|-------|-----------|-------------|-------------|-------------|
| 1     | 0<br>(0%) | 0<br>(0%)   | 0<br>(0%)   | 0<br>(0%)   |
| 2     | 0<br>(0%) | 8<br>(100%) | 0<br>(0%)   | 8<br>(89%)  |
| 3     | 0<br>(0%) | 0<br>(0%)   | 1<br>(100%) | 1<br>(11%)  |
| total | 0<br>(0%) | 8<br>(100%) | 1<br>(100%) | 9<br>(100%) |
|       | 1         | 2           | 3           | total       |

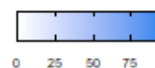
■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

3. 3rd Place

## Test Set

### Probability Heatmap

|    | 1  | 2  | 3  | Exp |
|----|----|----|----|-----|
| AH | 3  | 95 | 2  | 2   |
| AJ | 22 | 77 | 0  | 2   |
| G  | 39 | 61 | 0  | 2   |
| O  | 1  | 94 | 5  | 2   |
| P  | 6  | 93 | 1  | 2   |
| R  | 28 | 72 | 0  | 2   |
| U  | 18 | 81 | 0  | 2   |
| X  | 29 | 71 | 0  | 2   |
| AO | 0  | 10 | 89 | 3   |



3. 3rd Place

### NPA 6 SM Removed

formula

---

class ~ Dist\_(P,C)\_P + para\_SM

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy    McFadden\_R2

---

77.14%      0.407

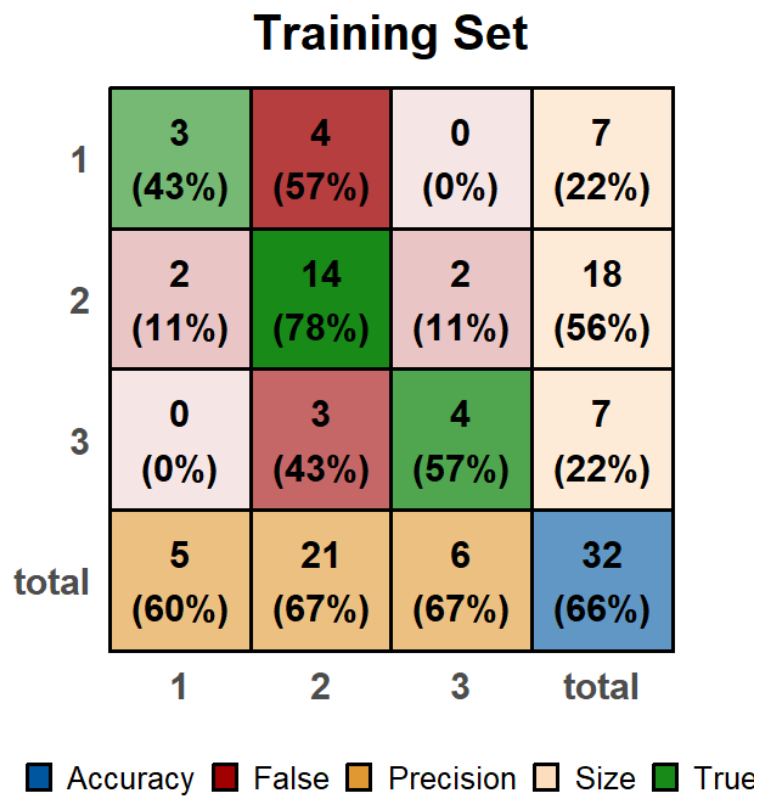
Model Coefficients

x

---

Dist\_(P,C)\_P    -134.422

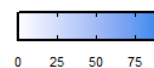
para\_SM            0.114



3. 3rd Place

Training Set  
Probability Heatmap

|    | 1  | 2  | 3  | Exp |
|----|----|----|----|-----|
| A  | 21 | 75 | 4  | 1   |
| AF | 19 | 77 | 4  | 1   |
| B  | 44 | 55 | 1  | 1   |
| C  | 64 | 35 | 1  | 1   |
| D  | 92 | 8  | 0  | 1   |
| E  | 23 | 74 | 3  | 1   |
| F  | 76 | 23 | 0  | 1   |
| AA | 31 | 66 | 2  | 2   |
| AB | 9  | 82 | 9  | 2   |
| AC | 5  | 79 | 17 | 2   |
| AE | 5  | 78 | 17 | 2   |
| AI | 1  | 36 | 63 | 2   |
| AK | 1  | 51 | 48 | 2   |
| AL | 21 | 75 | 4  | 2   |
| H  | 11 | 81 | 8  | 2   |
| I  | 31 | 67 | 2  | 2   |
| J  | 18 | 78 | 4  | 2   |
| L  | 3  | 71 | 26 | 2   |
| M  | 1  | 54 | 45 | 2   |
| S  | 83 | 17 | 0  | 2   |
| T  | 1  | 43 | 56 | 2   |
| V  | 12 | 81 | 7  | 2   |
| W  | 28 | 70 | 3  | 2   |
| Y  | 52 | 47 | 1  | 2   |
| Z  | 9  | 82 | 9  | 2   |
| AN | 1  | 45 | 54 | 3   |
| AP | 1  | 34 | 65 | 3   |
| AQ | 2  | 69 | 29 | 3   |
| AR | 2  | 65 | 33 | 3   |
| AS | 2  | 63 | 35 | 3   |
| AT | 1  | 34 | 65 | 3   |
| AU | 1  | 37 | 63 | 3   |



3. 3rd Place

## Test Set

|       |           |             |             |             |
|-------|-----------|-------------|-------------|-------------|
| 1     | 0<br>(0%) | 0<br>(0%)   | 0<br>(0%)   | 0<br>(0%)   |
| 2     | 0<br>(0%) | 8<br>(100%) | 0<br>(0%)   | 8<br>(89%)  |
| 3     | 0<br>(0%) | 0<br>(0%)   | 1<br>(100%) | 1<br>(11%)  |
| total | 0<br>(0%) | 8<br>(100%) | 1<br>(100%) | 9<br>(100%) |
|       | 1         | 2           | 3           | total       |

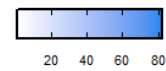
■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

3. 3rd Place

## Test Set

### Probability Heatmap

|    | 1  | 2  | 3  | Exp |
|----|----|----|----|-----|
| AH | 13 | 81 | 7  | 2   |
| AJ | 33 | 65 | 2  | 2   |
| G  | 23 | 74 | 3  | 2   |
| O  | 7  | 81 | 12 | 2   |
| P  | 6  | 81 | 13 | 2   |
| R  | 11 | 81 | 7  | 2   |
| U  | 9  | 82 | 10 | 2   |
| X  | 8  | 82 | 10 | 2   |
| AO | 1  | 34 | 65 | 3   |



3. 3rd Place

para SM Removed

formula

---

class ~ Dist\_(P,C)\_P + NPA\_6\_SM

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy    McFadden\_R2

---

54.29%    0.227

Model Coefficients

x

---

Dist\_(P,C)\_P    -54.923

NPA\_6\_SM    14.725

# Training Set

|       |            |             |            |             |
|-------|------------|-------------|------------|-------------|
| 1     | 2<br>(29%) | 5<br>(71%)  | 0<br>(0%)  | 7<br>(22%)  |
| 2     | 1<br>(6%)  | 15<br>(83%) | 2<br>(11%) | 18<br>(56%) |
| 3     | 0<br>(0%)  | 7<br>(100%) | 0<br>(0%)  | 7<br>(22%)  |
| total | 3<br>(67%) | 27<br>(56%) | 2<br>(0%)  | 32<br>(53%) |
|       | 1          | 2           | 3          | total       |

■ Accuracy
 ■ False
 ■ Precision
 ■ Size
 ■ True

3. 3rd Place

# Training Set

## Probability Heatmap

|    | 1  | 2  | 3  | Exp |
|----|----|----|----|-----|
| A  | 49 | 48 | 4  | 1   |
| AF | 38 | 56 | 5  | 1   |
| B  | 33 | 60 | 7  | 1   |
| C  | 41 | 54 | 5  | 1   |
| D  | 42 | 53 | 5  | 1   |
| E  | 51 | 46 | 3  | 1   |
| F  | 41 | 54 | 5  | 1   |
| AA | 34 | 60 | 7  | 2   |
| AB | 29 | 63 | 8  | 2   |
| AC | 20 | 68 | 13 | 2   |
| AE | 18 | 68 | 14 | 2   |
| AI | 32 | 61 | 7  | 2   |
| AK | 4  | 50 | 46 | 2   |
| AL | 6  | 57 | 37 | 2   |
| H  | 28 | 64 | 8  | 2   |
| I  | 9  | 65 | 26 | 2   |
| J  | 17 | 68 | 15 | 2   |
| L  | 9  | 64 | 28 | 2   |
| M  | 9  | 64 | 27 | 2   |
| S  | 2  | 35 | 63 | 2   |
| T  | 71 | 27 | 1  | 2   |
| V  | 31 | 61 | 7  | 2   |
| W  | 3  | 45 | 52 | 2   |
| Y  | 9  | 64 | 27 | 2   |
| Z  | 6  | 59 | 34 | 2   |
| AN | 7  | 60 | 33 | 3   |
| AP | 4  | 52 | 43 | 3   |
| AQ | 36 | 58 | 6  | 3   |
| AR | 4  | 51 | 44 | 3   |
| AS | 8  | 63 | 29 | 3   |
| AT | 8  | 63 | 29 | 3   |
| AU | 6  | 58 | 37 | 3   |



20 40 60

3. 3rd Place



## Test Set

|       |            |             |           |            |
|-------|------------|-------------|-----------|------------|
| 1     | 0<br>(0%)  | 0<br>(0%)   | 0<br>(0%) | 0<br>(0%)  |
| 2     | 1<br>(12%) | 7<br>(88%)  | 0<br>(0%) | 8<br>(89%) |
| 3     | 0<br>(0%)  | 1<br>(100%) | 0<br>(0%) | 1<br>(11%) |
| total | 1<br>(0%)  | 8<br>(88%)  | 0<br>(0%) | 9<br>(78%) |
|       | 1          | 2           | 3         | total      |

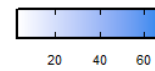
■ Accuracy 
 ■ False 
 ■ Precision 
 ■ Size 
 ■ True

3. 3rd Place

## Test Set

### Probability Heatmap

|    | 1  | 2  | 3  | Exp |
|----|----|----|----|-----|
| AH | 7  | 60 | 34 | 2   |
| AJ | 44 | 52 | 4  | 2   |
| G  | 46 | 50 | 4  | 2   |
| O  | 53 | 44 | 3  | 2   |
| P  | 18 | 68 | 14 | 2   |
| R  | 33 | 60 | 7  | 2   |
| U  | 26 | 65 | 9  | 2   |
| X  | 36 | 58 | 6  | 2   |
| AO | 8  | 63 | 28 | 3   |



3. 3rd Place